Dissertation

A Distributed Edge FLISR Solution & Network Simulation Test Platform

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Abstract

The energy sector is facing wide-scale changes with the rapid introduction of disruptive distributed energy sources, renewables, electric vehicles, and the changing relationship between the consumer and the utility. This has resulted in the need to leverage the capability of modern ICT and IoT technologies to meet the challenges presented by such changes. Of key concern is the resilience of the energy grid and the growing dependency on a stable supply to support business as usual activities and daily life for society at large. The following research explores how a novel grid resiliency strategy may be implemented, supported by the concept of Fault Identification, Isolation and Service Restoration, and enabled by Edge Computing techniques, to mitigate the effects of service loss to the end consumer and utility. This research focuses on the Irish energy grid in particular, exploring the background and motivation for such resiliency solutions, the related literature, research activities and insights gained from the results.

Declaration

I have read and understood the Departmental policy on plagiarism.

I declare that this document is my own work and has not been submitted in any form for another degree or diploma at any university or other institution of tertiary education.

Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

Signature ____________________________________

Darren Leniston

Date ____________________________________

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Glossary

Advanced Distribution Management System (ADMS) A software platform containing a suite of functions to support distribution network management and optimisation.

Customer Minutes Lost (CML) Measurement of the duration of interruption of supply over the course of a year. Consists of the average CML per consumer per year, in the event a supply interruption lasts up to or more than 3 minutes.

Distribution System Operator (DSO) Designated entity responsible for management of the distribution network from source of generation to final consumer.

Fault Passage Indicator (FPI) Indicator notifying that an on-pole switch has detected a fault on its specific point of the network.

Geographical Information System (GIS) Software system utilised to analyse and visualise data containing geographic information.

Load Break Fault Make Switch (LBFM) On-pole circuit breaker which can be controlled to energise or de-energise a section of network in the event of over-load, fault event or required maintenance.

Loss of Voltage Indicator (LVI) Indicator notifying that an on-pole switch has lost supply on its specific point of the network.

Low Voltage (LV) Network Distribution network which transmits to end customers, generally at mains rated voltage (230V single phase & 400V 3 phase for the Irish LV network).

Medium Voltage (MV) Network Distribution network which transmits at voltages above 1kV (kilovolt) (10kV, 20kV & 38kV for the Irish MV network).

List of Acronyms

CI Customer Interruption

DA Distribution Automation

FLISR Fault Location, Isolation and Service Restoration

ICT Information and Communication Technology

IGSD Intelligent Grid Switching Devices

IoT Internet of Things

KPI Key Performance Indicator

MAS Multi Agent Approach

1 Introduction

With the rapid changes as a result of the integration of advanced ICT & IoT technologies into business as usual and daily activities, this has lead to large bodies of research in an effort to maximise the potential these technologies and their insights can unlock. Likewise, the energy sector has seen rapid changes with the introduction of renewable energy sources, electric vehicles, and a drive to utilise software based solutions which leverage embedded devices. This drive towards the smart grid forms the backdrop of the following dissertation, in particular, this research focuses on the aspect of energy grid resiliency. This dissertation expands on the learnings from previous related research as part of a European Horizon 2020 project [17], to develop a novel approach to an automated grid resilience strategy in the context of the Irish energy grid.

1.1 Document overview

To streamline the review of the final version of this document, this section outlines the additions and changes from the interim document. Both Section 1 & Section 2 remain predominantly unchanged from the interim report submission, aside from minor edits to reframe the language in context of the final submission. Content in Section 3 has been expanded to include detailed descriptions of the process of analysing the topology data, the FLISR Edge solution architecture, descriptions of the trial networks, data formats and the interaction of the FLISR algorithms with physical assets. The experiments outlined in Section 3.3 now include detailed descriptions of the tests conducted with the network simulation platform and FLISR algorithms, how the components operate and results are captured. The results of the experiments are discussed in Section 4 and how they relate to the research objectives, also detailed are the findings, the relevance of the results to the DSO and consumer, and proposes future enhancements. The conclusions gleaned from the research activities are described in the Section 5, the exploitation activities outlined in Section 6 remain unchanged from the interim report.

1.2 Research background

The energy grid is of vital importance in today's society and is a key piece of the infrastructure which enables activities across a range of industries, services and domestic settings. Like many aspects of the modern world then, the energy industry has been seeing a paradigm shift in the last number of years with the introduction of state-of-the-art ICT and IoT technologies in an effort to modernise the energy sector; in terms of production, transmission and delivery to consumers [11]. These new technologies are taking the form of smart sensors and power monitoring at both the grid and domestic level, unlocking new data and granting greater insight into the state of the grid [3]. This, combined with controllable devices, is leading to Advanced Distribution Management Systems (ADMS), enabling a fine-grained view and greater control of the energy grid.

There are a number of factors driving this race to modernise the grid. While traditional energy infrastructures operate on a top-down model, where supply to consumers from the transmission system follows a uni-directional flow and energy generation throughout the grid is more centralised [19], this model of energy generation and supply is transforming at a rapid pace. With the introduction of distributed and renewable sources of energy production, electric vehicles, microgrids and the changing relationship between the consumer and the utility, whereby certain consumers are becoming "prosumers" (i.e., generating and selling their excess generation back to the grid) the structure of the energy grid is evolving [12]. In this way, the modernisation of the grid through ICT and IoT technologies intends to meet the challenges posed by the disruptive nature of these changes.

Within this context, the research pursued in this document aims to meet the requirement for a resilient energy grid, utilising modern technologies and techniques, to reduce and mitigate the effects of disruptive events on the

grid and their impact on the consumer and utility. To meet this goal, the research leverages the concept of Fault Location, Isolation and Service Restoration (FLISR), a grid resilience strategy which in the event of a fault on the grid, aims to automate the identification of said fault, isolating physically damaged sections of the network and restoring service to consumers on lines affected by loss of service but are otherwise undamaged.
In combination with the FLISR technique, this research also explores the application of Edge Computing and novel representations of grid topology data to realise a distributed, fault-tolerant FLISR solution which is not fully reliant on a centralised cloud deployment, by moving subsets of the grid topology and FLISR logic to low-cost edge devices installed on existing grid infrastructure. In order to test, evaluate and demonstrate the solution, a simulation platform was also developed. This research is framed in the context of the Irish distribution grid, in particular three distribution grids of varying topologies and features located throughout the country.

1.3 Motivations for the research

In tandem with the rapid transformation of the energy sector, the need for a resilient grid is vital with the growing reliance on consistent power delivery for societal, business and manufacturing activities. Coupled with this is the rising concern of climate change and its impact on everyday life. Recent times have seen an increase in the number and severity of weather events which cause localised and wide-scale damage to infrastructure. Such events often lead to loss of energy supply to consumers, businesses and other institutions through damage caused to wires, transformers, substations, equipment and other grid assets [24]. In relation to the Irish energy network, weather events such as ex-hurricane Ophelia in 2017 resulted in 5,500 damaged overhead lines and 385,000 homes & businesses without power [13].

In addition, the Distribution System Operator (DSO) in Ireland, ESB Networks, is faced with fines by the regulator based on specific KPIs which are directly influenced by the amount of unplanned service interruptions and loss of supply, namely Customer Interruption (CI) & Customer Minutes Lost (CML). For reference, Ophelia resulted in a total of 665.6 million CML, culminating in a 4.86 million euro fine imposed by the Irish energy regulator [13]. However, it must be said that although Ophelia was a rare event, 2017 was not an outlier in terms of total CML per customer, as detailed in ESB Networks 2020 report [21] and illustrated in Figure 1.

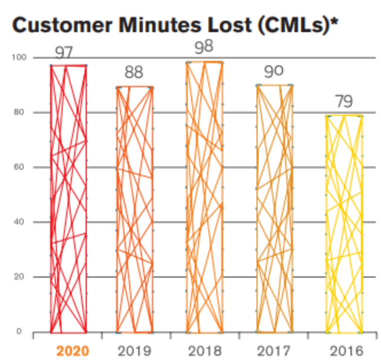


Figure 1: CML figures 2016-2020

In this way then, techniques such as FLISR offer the possibility to drastically mitigate against the effects of inclement weather events. However, there are limitations to the DSO in regards to deploying a FLISR solution to the grid, such as the cost of specialised equipment, in addition to the installation and requirement for large-scale software platforms such as ADMS. In combination with this, weather events can also affect communication networks, thereby reducing or even negating the benefit of a cloud-based or centralised FLISR implementation. With this factor in mind, the concept of edge computing is one that may greatly benefit the operation of a FLISR algorithm in the field, by moving a subset of the business logic and data processing away from a centralised system, to the edge of the network. In the context of FLISR, this would enable sections of the network to operate independently if needed to isolate and restore fault events, providing a fault-tolerant FLISR solution that is able to operate effectively in conditions such as Ophelia.

1.4 Research objectives

The key objectives which framed the direction of the research activities centered around delivering a novel FLISR solution, leveraging modern technologies which can operate both centrally and in a distributed fashion at the network edge, and to test, evaluate and demonstrate the operation and effectiveness of the solution through the use of a simulation platform, utilising a number of grid topologies to assess the FLISR algorithms effectiveness against a variety of network configurations located throughout Ireland.

The main output of the research is gauging how the developed FLISR solution may reduce CML in real-world operation, to that end, the following questions guided the research activities.

- **R1 - How effective will the outputs of the FLISR solution be in terms of reducing DSO fault KPIs?**

This question is crucial in regards to evaluating the value of the research activities and FLISR solution implemented, both compared to alternative approaches to the FLISR concept undertaken in other research, and in relation to the tested grid topologies and extrapolated to the Irish energy grid as a whole.

- **R2 - Given a similar set of conditions, how will the results produced by the centralised FLISR algorithm compare to the distributed FLISR algorithms operating locally at the edge of the network?**

A key feature of the distributed FLISR algorithm is to be in itself fault-tolerant, and therefore able to operate and produce results locally in the event that communication with the central, “master” FLISR algorithm is unavailable. Given the limited view of the network granted to the FLISR algorithms operating locally at the edge, in comparison to the overall system awareness of the central FLISR algorithm, will the results produced by both be similar and, crucially, valid.

- **R3 - Given a diverse set of network topologies, will the FLISR solution output valid results for a variety of grid configurations?**

For the FLISR solution to be effective, a key requirement will be to operate and produce valid results given a wide set of grid topologies with varying network configurations in terms of structure, number of nodes and device types.

2 Review of the literature

The following section explores and reviews a number of items of literature related to the aforementioned research topics. The findings of this review are collated by a number of sub-themes; digitalisation of the grid, FLISR and edge computing, which directly relate to the concepts and technologies applied to the distributed edge FLISR solution.

2.1 Digitalisation of the grid

There is a wide range of literature available on the concept of the smart grid, and what this digitalisation and modernisation of the energy network means for the energy industry. *Fang et al.* [11] describe a smart grid as utilising two-way flows of both electricity and data in an aim to create an automated and distributed advanced energy delivery network, and through the use of modern ICT lead to a more efficient and flexible grid which can respond to a variety of events. The article in question describes the requirements for the smart grid from the perspective of three major systems; *smart infrastructure systems*, *smart management systems* and *smart protection systems*. Of most relevance here, are smart infrastructure systems, which are described as supporting two-way data flows via wired and wireless communication methods, and smart protection systems of which failure diagnosis and grid self-healing are a key factor.

An article by *Güngör et al.* [15] outlines the trends resulting in the need for this new, smarter energy infrastructure, taking the US as an example, reporting demand and consumption of electricity has increased by 2.5% annually over 20 years. This growing demand, in combination with factors such as greenhouse emissions, the environmental impact of extracting fossil-fuels for energy production and their growing scarcity, is driving the need for renewable sources and more flexibility in how the energy grid operates by enabling prosumer activities, microgrids, virtual power plants and increased grid resiliency to meet these challenges.

The paper describes the smart grid as a "modern electric power grid infrastructure for improved efficiency, reliability and safety, with smooth integration of renewable and alternative energy sources, through automated control and modern communication technologies". In addition, they describe how the reliability of the energy infrastructure and supply can be supported and maintained in the smart grid, and how the impact of equipment failure, capacity constraints and natural accidents which lead to supply interruption and outages, can be mitigated by online power system monitoring, diagnostics and protection.

As described in the referenced articles, enabling the smart grid requires reliable information and data flow from a variety of smart devices both in the field and in the domestic setting, therefore the communication methods utilised are of key concern. In a separate article, *Güngör et al.* [14] discuss how wireless communication methods can meet some of the challenges presented by the move to the smart grid. Within the paper, it is explained that while traditionally power system monitoring was realised through wired communication via cabling from pole to pole, this method is expensive and therefore not widely implemented today due to its financial impact.

Related to these communication requirements, *Priya* & *Malhotra* [25] describe how the need for efficient, reliable communication on the smart grid can be achieved by leveraging the capabilities of the telecommunication network, specifically referencing 5G and its fast, low-latency features and how these can meet the challenge of capturing large amounts of data from the field and enable quick, automated control services to manage, balance and ensure the resilience of the grid.

2.2 Fault Location, Isolation & Service Restoration

With resiliency of the network a key concern to the DSO and frequently mentioned as a requirement for the smart grid in the literature, FLISR being one such technique which has been explored and discussed in various research. *Agüero* [2] describes FLISR as a self-healing technique, though does make mention of certain authors preferring the term "self-restoration", as in reality no system can currently fix physical damage to the grid without direct

intervention by DSO personnel. The paper discusses how Distribution Automation (DA) technologies can enable remote monitoring, coordination and operation of distributed network assets at substation, feeder and customer level. Most relevant, is the description of utilising remote, on-pole switching devices as a method of sectionalising and restoring supply to customers in the event of a fault. The paper describes how the need for manual investigation and operating of switching devices on the part of DSO operation staff can be reduced or rendered redundant through automated FLISR, and the benefits of such a solution being the improvement of reliability statistics and the savings made due to the reduced cost of outages.
However, the text posits that leveraging such devices for self-healing is more appropriate for urban and suburban areas, rather than remote rural locations, where radial distribution is more commonplace (i.e., one source of power for consumers on the network). Though it could be argued that resilience strategies such as FLISR are more necessary for rural areas, due to the increased prevalence of faults as a result of deterioration and damage in remote locations.

A report produced by the US Department of Energy in 2014 [23] aimed to gather metrics on five separate FLISR solutions implemented throughout the United States over the course of one year. The findings included a large reduction in the number of customers affected by service loss (270,000) with 38 million fewer CML in total, translating to a reduction of 45% in the amount of interruptions and 51% in CML per fault event. The report specifies that an essential component for successful FLISR operations are the communications networks used to monitor and enable control over switching systems and devices, in combination with suitable interfaces which allow interoperability between various DSO systems and device types on the grid.
Each FLISR system explored utilises various device types, communication methods, modes of operation and location of the deployed system as part of its implementation. For example, a utility named CenterPoint trialed their FLISR system, named “Self-Healing Grid” which is supported by Intelligent Grid Switching Devices (IGSD) and requires manual validation by a System Operator in order to enact control operations. Conversely, a utility named Duke leveraged electronic reclosers, circuit breakers and line sensors for its FLISR system, dubbed “Self-Healing Teams” to enable a fully-automated solution.
The report specifies a number of learning’s gathered by the utilities from the FLISR trials, such as the need for the telecommunications network to be more resilient than the power delivery systems which rely upon them, and the necessity for the utility to consider the additional steps required to successfully deploy and operate a FLISR system. Steps identified included the testing and installation of new equipment, and the frequent firmware and software updates necessary to ensure the system continues to operate effectively due to its autonomous nature, though one utility, Georgia Power, found that developing a simulator to test and validate FLISR eliminated the need for extensive field trials and served as a training tool for operation staff.

A paper by *Eriksson et al.* [10] further discusses the limitations and considerations when implementing FLISR solutions, specifically centralised deployments, such as having a single point of failure and the requirement for the handling of large amounts of real-time data, large network topologies and the processing necessary to ensure consistent operation. They propose a distributed FLISR utilising a Multi Agent Approach (MAS) devices, tested against a radial network commonly found in remote areas, and applying Prim’s Algorithm to devise the FLISR logic. The solution proposed was tested against a cyber-physical testbed, with the results finding that the MAS approach to FLISR provided efficient grid stabalisation and fault restoration.

Most relevant is a report produced by ESB Networks as an output of the H2020 project SOGNO [20]. This project focused on delivering turn-key grid services for power quality monitoring, estimating network state, load prediction and forecasting as well as FLISR. The FLISR implementation described in the report leveraged communication from Load Break Fault Make (LBFM) on-pole circuit breakers in the field. In the event of a fault, these switches communicate a Fault Passage Indicator (FPI) and Loss of Voltage Indicator (LVI) and in this way it is possible to identify a fault occurrence. ESB Networks notes that a number of their networks currently rely upon manual intervention on the part of control room operation staff to manually identify and resolve fault events when they occur. This need for human intervention greatly increases the time taken to restore supply to consumers, leading to higher CML per event. While the FLISR solution tested as part of the SOGNO project produced good results, ESB Networks noted that its reliance on a centralised instance and telecommunications to operate is a severe limitation.

2.3 Edge computing

Over the last number of years, cloud computing has provided a method for large-scale data collection and processing for multiple use-cases and industries, however this dynamic is changing with the proliferation of IoT.
Shi et al. [26] describe this as the post-cloud era, where embedded IoT will not only serve as a data source but also consume and act on this data at the edge. As more data sources come online, the level of processing required by cloud computing services to collect, parse and act on this data increases. This, coupled with network bandwidth constraints when considering the large amounts of data gathered from distributed sources means new solutions are necessary and the researchers propose the edge computing concept as a solution to these challenges. The literature defines edge computing as enabling technologies which allow computing activities to be performed at the edge of the network on behalf of cloud-based services, therefore working as a data consumer as well as a data producer. One such example is the smartphone, which can serve as both a producer of data for cloud-based services and as a processing device at the edge. An example use-case being health-related applications, where user metrics are collected and processed on the device, and resulting health data is passed and stored on a cloud-based service.

Deploying and managing edge resources is a key factor when considering edge computing solutions, *Taleb et al.* [27] describe the importance of orchestration frameworks to enable integration and management of edge devices in terms of resource allocation and service management in regards to software and firmware updates. The paper describes how utilising virtualisation, such as containerised software services, supports the packaging, delivering, deploying and updating of software services operating on edge computing devices located in the field.
In relation to the smart grid, a paper by *Zhang et al.* [30] proposes that a number of challenges facing the smart grid, such as high computational cost and transmission delay can be met through the application of edge computing. Similarly, *Chen et al.* [6] discuss the benefits of edge computing in supporting smart grid operations, with specific reference to grid resiliency and self-healing, describing how the dependency on centralised and cloud-based systems may lead to less efficient operation of grid control and monitoring services. Furthermore, the proximity of edge computing devices to grid assets may reduce latency when intelligent decisions are made locally, and provide finer-grained location awareness and monitoring of the network for the DSO, this is particularly important at the MV and LV levels where visibility of the grid from a system perspective is greatly reduced.

2.4 Summary

The exploration of the literature in the related thematic areas, highlights the level of research undertaken to realise the smart grid and the challenges faced in doing so. In particular, resiliency of the grid is referenced frequently as a key goal, with FLISR providing an effective method of achieving this objective. In regards to FLISR, there are many applications in the research that utilise various methods of fault identification and resolution in terms of the algorithms applied, communication protocols, standards and methods of control, such as fully autonomous FLISR as well as those that require human intervention. A key concern identified in the literature is the need to ensure such grid resiliency solutions are in themselves fault-tolerant and therefore not dependent on centralised systems or core telecommunication networks. In reviewing the research, distributed FLISR solutions appear to be a means of resolving this limitation.

In realising a distributed FLISR solution edge computing emerges as an ideal paradigm to implement this functionality. By removing the need for large data processing and decision making functionality from cloud-based systems and moving subsets of this logic to the edge of the network, grid resiliency solutions, such as distributed FLISR, should benefit from reduced latency and dependency on core systems in order to operate effectively. Indeed, the literature describes how edge computing may be a key component in realising the smart grid, due to the amount of data generated and computation required to effectively manage the energy distribution system. In combination with this, is the difficultly in effecting large scale changes on the core distribution grid due to expense and dealing with legacy components. Therefore, enabling more computation, data collection and control at the edge of the network may afford smart grid functionality, implementation of concepts such as the VPP, microgrids and prosumer activities.

3 Methods

The following section details the development of the FLISR Edge solution, describing the development methodology, underlying network topology and how key data points were identified. Also discussed is the architecture defined to realise the network simulation platform and FLISR Edge algorithms, the data formats and interfaces required to enable compatibility with on pole circuit breaks and how the FLISR algorithms ingest and determine operational steps in response to fault events.

3.1 Formalisation of stakeholders & development process

The research work applies an iterative, agile model, in particular employing the concept of Behavior Driven Development (BDD) which aims to identify and define system functionality from the lens of key stakeholders and defines functions and business logic as "features" which guide the development process [1]. A system has one or more stakeholders, these are the actors outside of the system that interact with the application in question. Actors will have interests, goals or concerns in relation to that system, and may take the form of a person, outside system, subsystem or device to name a few. These actors will have various roles in regards to the operation of the system, goals which the system aims to fulfill and these are defined prior to development in line with the following format.

- **Actor Name:** A clear identifier for the actor.
- **Actor Type:** One of person, device, system etc.
- **Actor Description:** Detailed description of the actor.
- **Actor Goal:** What the actor aims to achieve with the system.

It was necessary to define users for both the simulation platform and the FLISR Edge algorithm itself, presented in Table 1, and these served as the initial starting point to then define user stories which guided the development tasks. The individual user stories for each actor can be found in Appendices A & B.

Table 1: FLISR Edge Simulation Platform & Algorithm Users

Actor name	Actor type	Actor description	Actor Goal
DSO (Distribution System Operator)	Organisation (Utility)	Network owner & operator	They want a FLISR solution to reduce the level of CML caused by fault events on the distribution grid
CRO (Control Room Operator)	Person (DSO personnel)	Asset operator	They want to simulate fault events to capture FLISR Edge outputs to evaluate their effectiveness in the field
FLISR Master	System	FLISR centralised algorithm & distributed system orcestrator	Overall FLISR Edge system management, provisioning distributed algorithms to network edge, centralised algorithm with full network view
FLISR Node	Subsystem	Distributed FLISR Edge device	Subsystem which runs at the network edge on an individual switch, it will act independently if a connection to the central system is unavailable

The defined user stories can then be translated into a number of implementable system features, which are then broken down further into a array of development tasks. Within the context of FLISR Edge, GitHubs "Projects" space was leveraged to provide a task board which was populated with separate issue tickets which relate to a specific research, development or documentation task and a separate GitHub repository was provisioned for each entity of the FLISR Edge system. All features implemented are version controlled through separate branches, commits and were tested and verified before merging in order to avoid breaking changes and to keep the main branch clean. Additionally, Wiki spaces were utilised to capture documentation and research items throughout development to be included in the final research document.

3.2 Analysis & import of network topology

As the underlying grid topologies support both the FLISR algorithms and the simulation platform, extracting this information from the available data served as a starting point for development. The data in question is derived from a Geographical Information System (GIS) named Synergy, this system maps the distribution network through the use of geospatial data [16] in addition to metadata related to each individual asset, whether this entity is a pole, transformer, feeder, switch, substation or other asset type. The properties contained within this metadata depend on the asset type, for example which nodes (poles) are connected, their phases (RST) which denote whether they are part of the three phase (MV) or single phase (LV) network, which poles are equipped with Load Break Fault Make (LBFM) or Recloser circuit breakers and which of these are denoted as “Normally Open” and therefore serve as the boundary between network lines.

The work of transforming this data into a usable state within a graph database format, and identifying key properties consisted of four primary steps; firstly, mapping the base Synergy data from a relational format to neo4j, analysing the key properties to include in the final dataset, determining a method of approximating the number of connected customers to each node and finally correlating the identified properties into a consistent model.

3.2.1 Mapping of relational to graph database format

Prior to analysis of the data, the export from Synergy which takes the form of a “MDB” (Microsoft Access Database) file was converted and ingested into a Postgres database, from here it was possible to generate an Entity Relationship Diagram (ERD), presented in Figure 3, to obtain a clearer view of the assets included in the data and their associated properties. The steps to translate the base MDB files to a relational format is based on previous research work [17], however streamlined to allow for simplified ingestion of additional distribution grids in future research activities.

From the analysis of the ERD it is clear that their is a deeply coupled relationship between entities in this data, and one of the goals of the research is to represent this data in a novel format through leveraging the concept of Graph Databases, which focuses on the connections between data and affords a more ontological view [28] of the relationship between entities. To achieve this aim, the Synergy data was ingested into a “neo4j” graph database in a containerised environment as illustrated in Figure 2.

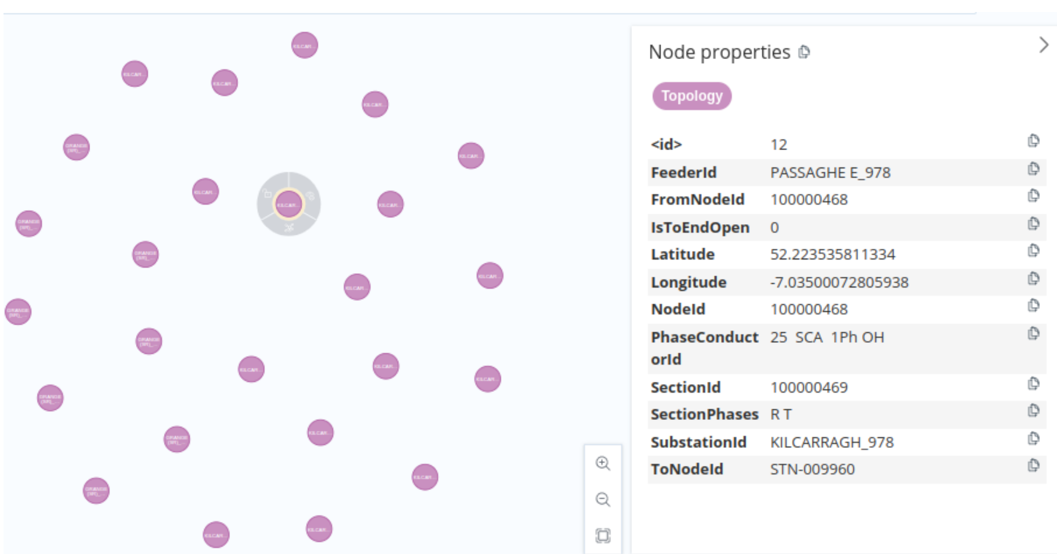


Figure 2: Topology data ingested in graph database

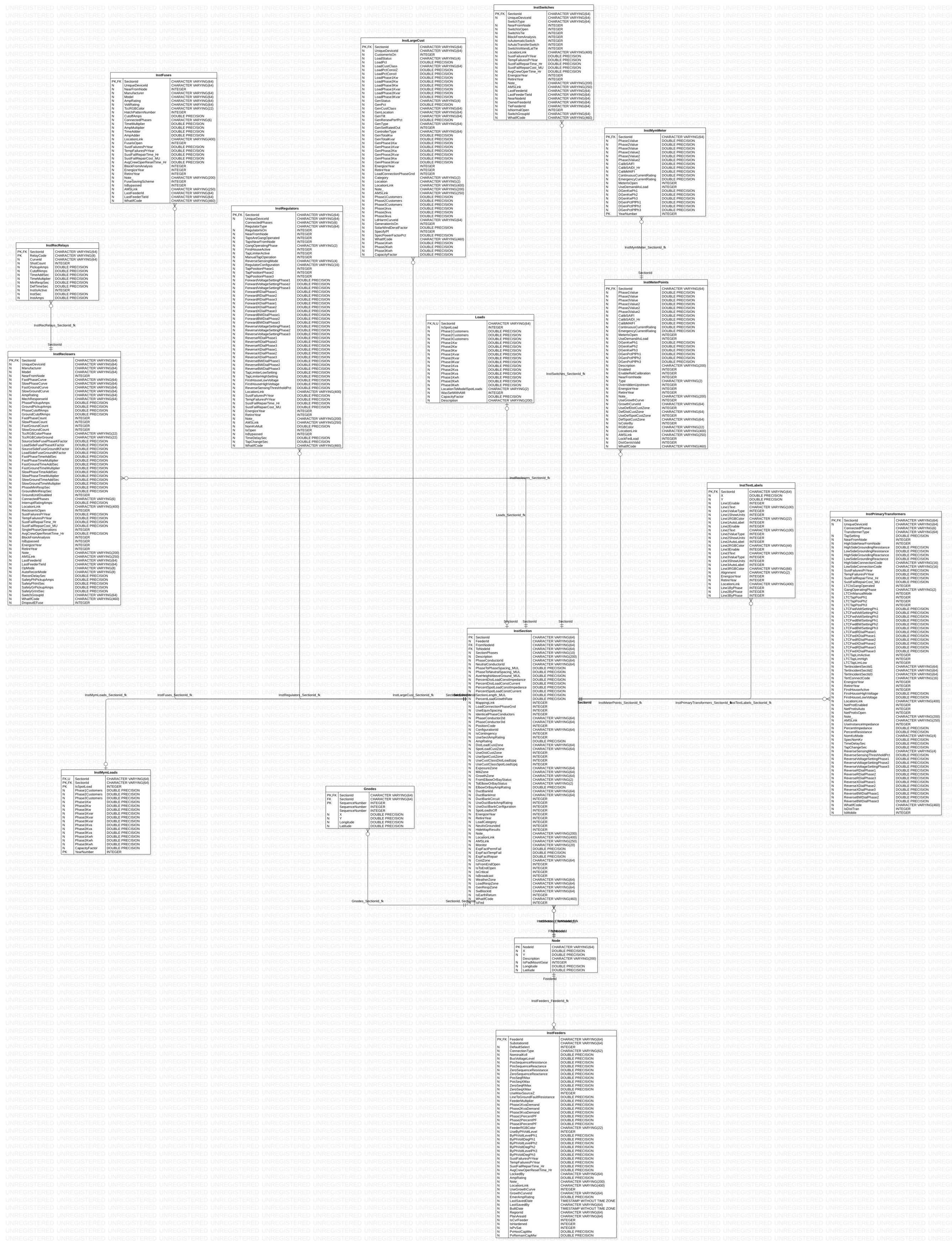


Figure 3: Synergy data ERD

3.2.2 Identification of key node properties

The Synergy dataset contains tables which are either empty or contain information which does not provide any value when considering a FLISR implementation, and to this end the dataset was explored thoroughly to discern useful data considering relevant assets and to organise it in such a way that it could be efficiently loaded into the simulation platform and ingested by the FLISR algorithms.

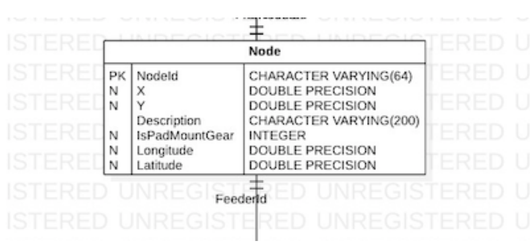
A key table identified "InstSections" forms the hub of the data schema and maps to a majority of other entities via the property "SectionId". In the context of the Synergy system individual poles are referred to as "nodes", and therefore the properties "FromNodeId" and "ToNodeId" are pivotal in determining how poles are connected to one another, additionally the property "SectionPhases" specifies whether the specific section is part of the MV or LV network, Figure 4 details the full "InstSection" schema.

InstSection

PK	SectionId	CHARACTER VARYING(64)
N	FeederId	CHARACTER VARYING(64)
FK	FromNodeId	CHARACTER VARYING(64)
FK	ToNodeId	CHARACTER VARYING(64)
N	SectionPhases	CHARACTER VARYING(10)
N	Description	CHARACTER VARYING(200)
N	PhaseConductorId	CHARACTER VARYING(64)
N	NeutralConductorId	CHARACTER VARYING(64)
N	PhaseToPhaseSpacing_MUL	DOUBLE PRECISION
N	PhaseToNeutralSpacing_MUL	DOUBLE PRECISION
N	AveHeightAboveGround_MUL	DOUBLE PRECISION
N	PercentDistLoadConstImpedance	DOUBLE PRECISION
N	PercentDistLoadConstCurrent	DOUBLE PRECISION
N	PercentSpotLoadConstImpedance	DOUBLE PRECISION
N	PercentSpotLoadConstCurrent	DOUBLE PRECISION
N	SectionLength_MUL	DOUBLE PRECISION
N	PercentLoadGrowthRate	DOUBLE PRECISION
N	MappingLink	INTEGER
N	LoadConnectionPhaseGnd	INTEGER
N	UseEquivSpacing	INTEGER
N	IdenticalPhaseConductors	INTEGER
N	PhaseConductor2Id	CHARACTER VARYING(64)
N	PhaseConductor3Id	CHARACTER VARYING(64)
N	PositionCode	INTEGER
N	ConfigurationId	CHARACTER VARYING(64)
N	IsContingency	INTEGER
N	UseSectAmpRating	INTEGER
N	AmpRating	DOUBLE PRECISION
N	DistLoadCustZone	CHARACTER VARYING(64)
N	SpotLoadCustZone	CHARACTER VARYING(64)
N	UseDistCustZone	INTEGER
N	UseSpotCustZone	INTEGER
N	UseCustClassDistLoadIzpq	INTEGER
N	UseCustClassSpotLoadIzpq	INTEGER
N	ExposureZone	CHARACTER VARYING(64)
N	MitiZone	CHARACTER VARYING(64)
N	GrowthZone	CHARACTER VARYING(64)
N	FromElbowOrBayStatus	CHARACTER VARYING(2)
N	ToElbowOrBayStatus	CHARACTER VARYING(2)
N	ElbowOrBayAmpRating	DOUBLE PRECISION
N	DuctBankId	CHARACTER VARYING(64)
N	DuctBankInst	CHARACTER VARYING(64)
N	DuctBankCircuit	INTEGER
N	UseDuctBankAmpRating	INTEGER
N	UseDuctBankConfiguration	INTEGER
N	SpotLoadIsOff	INTEGER
N	EnergizeYear	INTEGER
N	RetireYear	INTEGER
N	LoadCategory	INTEGER
N	NeutIsGrounded	INTEGER
N	HideMapResults	INTEGER
N	Note_	CHARACTER VARYING(200)
N	LocationLink	CHARACTER VARYING(400)
N	AMSLink	CHARACTER VARYING(250)
N	Monitor	CHARACTER VARYING(20)
N	ExpFactPermFail	DOUBLE PRECISION
N	ExpFactTempFail	DOUBLE PRECISION
N	ExpFactRepair	DOUBLE PRECISION
N	CostZone	CHARACTER VARYING(64)
N	IsFromEndOpen	INTEGER
N	IsToEndOpen	INTEGER
N	IsCritical	INTEGER
N	IsBroadcast	INTEGER
N	WeatherZone	CHARACTER VARYING(64)
N	LoadRespZone	CHARACTER VARYING(64)
N	GenRespZone	CHARACTER VARYING(64)
N	SwBlockId	CHARACTER VARYING(64)
N	IsEarthReturn	INTEGER
N	WhatIfCode	CHARACTER VARYING(460)
N	IsFed	INTEGER

Figure 4: InstSection Schema

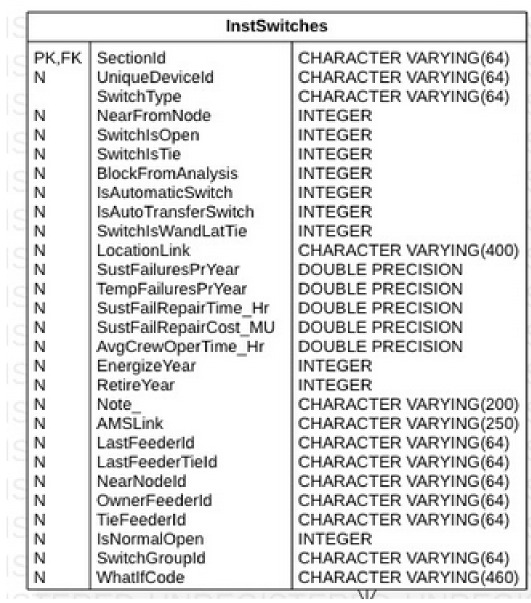
Geolocation data for individual nodes is contained within the "Nodes" table, in both Cartesian (X & Y) and Geographic (latitude & longitude) coordinate formats as shown in Figure 5, however it was noted that the latitude and longitude properties had been populated with the X & Y coordinates, and while there is a level of support for Cartesian coordinates within many software tools and packages for mapping objects, the vast majority utilise geographic coordinates as standard [4]. To mitigate this factor, the X & Y coordinates were converted to their corresponding latitude & longitude coordinates as part of the conversion process from the original MDB database format to Neo4j.



Node		
PK	NodeId	CHARACTER VARYING(64)
N	X	DOUBLE PRECISION
N	Y	DOUBLE PRECISION
	Description	CHARACTER VARYING(200)
N	IsPadMountGear	INTEGER
N	Longitude	DOUBLE PRECISION
N	Latitude	DOUBLE PRECISION

Figure 5: Node Schema

Nodes with LBFM circuit breakers and other asset types equipped which fall into this bracket are contained within the "InstSwitches", which contains a number of properties as illustrated in Figure 6, most notably "SectionId" and "UniqueDeviceId" for linking to the "InstSection" and "Node" entities, in addition to "SwitchType" where LBFM circuit breakers are denoted as "Soules" and "SwitchIsOpen" which specifies whether the asset is configured to be a normally open point.



InstSwitches		
PK,FK	SectionId	CHARACTER VARYING(64)
N	UniqueDeviceId	CHARACTER VARYING(64)
	SwitchType	CHARACTER VARYING(64)
N	NearFromNode	INTEGER
N	SwitchIsOpen	INTEGER
N	SwitchIsTie	INTEGER
N	BlockFromAnalysis	INTEGER
N	IsAutomaticSwitch	INTEGER
N	IsAutoTransferSwitch	INTEGER
N	SwitchIsWandLatTie	INTEGER
N	LocationLink	CHARACTER VARYING(400)
N	SustFailuresPrYear	DOUBLE PRECISION
N	TempFailuresPrYear	DOUBLE PRECISION
N	SustFailRepairTime_Hr	DOUBLE PRECISION
N	SustFailRepairCost_MU	DOUBLE PRECISION
N	AvgCrewOperTime_Hr	DOUBLE PRECISION
N	EnergizeYear	INTEGER
N	RetireYear	INTEGER
N	Note_	CHARACTER VARYING(200)
N	AMSLink	CHARACTER VARYING(250)
N	LastFeederId	CHARACTER VARYING(64)
N	LastFeederTieId	CHARACTER VARYING(64)
N	NearNodeId	CHARACTER VARYING(64)
N	OwnerFeederId	CHARACTER VARYING(64)
N	TieFeederId	CHARACTER VARYING(64)
N	IsNormalOpen	INTEGER
N	SwitchGroupId	CHARACTER VARYING(64)
N	WhatIfCode	CHARACTER VARYING(460)

Figure 6: InstSwitches Schema

While Reclosers are denoted as circuit breakers, within the Synergy dataset information regarding these assets is assigned its own entity named "InstReclosers" with a number of properties associated as described in Figure 7. Similar to the previous entity, the "InstReclosers" table contains the "SectionId" and "UniqueDeviceId" properties. Where it differs is the inclusion of "Manufacturer" and "Model", in this specific instance, the assets of interest are denoted as "Nu-Lec" model "N Series", in addition, those Recloser circuit breakers configured as normally open are specified by the "RecloserIsOpen" property.

InstReclosers		
PK,FK	SectionId	CHARACTER VARYING(64)
N	UniqueDeviceId	CHARACTER VARYING(64)
N	Manufacturer	CHARACTER VARYING(64)
N	Model	CHARACTER VARYING(64)
N	NearFromNode	INTEGER
N	FastPhaseCurve	CHARACTER VARYING(64)
N	SlowPhaseCurve	CHARACTER VARYING(64)
N	FastGroundCurve	CHARACTER VARYING(64)
N	SlowGroundCurve	CHARACTER VARYING(64)
N	AmpRating	CHARACTER VARYING(64)
N	MechResponseId	CHARACTER VARYING(64)
N	PhasePickupAmps	DOUBLE PRECISION
N	GroundPickupAmps	DOUBLE PRECISION
N	PhaseCutoffAmps	DOUBLE PRECISION
N	GroundCutoffAmps	DOUBLE PRECISION
N	FastPhaseCount	INTEGER
N	SlowPhaseCount	INTEGER
N	FastGroundCount	INTEGER
N	SlowGroundCount	INTEGER
N	TccRGBColorPhase	CHARACTER VARYING(22)
N	TccRGBColorGround	CHARACTER VARYING(22)
N	SourceSideFusePhaseKFactor	DOUBLE PRECISION
N	LoadSideFusePhaseKFactor	DOUBLE PRECISION
N	SourceSideFuseGroundKFactor	DOUBLE PRECISION
N	LoadSideFuseGroundKFactor	DOUBLE PRECISION
N	FastPhaseTimeAddSec	DOUBLE PRECISION
N	FastPhaseTimeMultiplier	DOUBLE PRECISION
N	FastGroundTimeAddSec	DOUBLE PRECISION
N	FastGroundTimeMultiplier	DOUBLE PRECISION
N	SlowPhaseTimeAddSec	DOUBLE PRECISION
N	SlowPhaseTimeMultiplier	DOUBLE PRECISION
N	SlowGroundTimeAddSec	DOUBLE PRECISION
N	SlowGroundTimeMultiplier	DOUBLE PRECISION
N	PhaseMinRespSec	DOUBLE PRECISION
N	GroundMinRespSec	DOUBLE PRECISION
N	GroundUnitDisabled	INTEGER
N	ConnectedPhases	CHARACTER VARYING(6)
N	InterruptRatingAmps	DOUBLE PRECISION
N	LocationLink	CHARACTER VARYING(400)
N	RecloserIsOpen	INTEGER
N	SustFailuresPrYear	DOUBLE PRECISION
N	TempFailuresPrYear	DOUBLE PRECISION
N	SustFailRepairTime_Hr	DOUBLE PRECISION
N	SustFailRepairCost_MU	DOUBLE PRECISION
N	SinglePhaseOperations	INTEGER
N	AvgCrewOperResetTime_Hr	DOUBLE PRECISION
N	BlockFromAnalysis	INTEGER
N	IsBypassed	INTEGER
N	EnergizeYear	INTEGER
N	RetireYear	INTEGER
N	Note_	CHARACTER VARYING(200)
N	AMSLink	CHARACTER VARYING(250)
N	LastFeederId	CHARACTER VARYING(64)
N	LastFeederTieId	CHARACTER VARYING(64)
N	OpMode	CHARACTER VARYING(8)
N	AutoReconMode	CHARACTER VARYING(8)
N	ReconDelaySec	DOUBLE PRECISION
N	SafetyPhPickupAmps	DOUBLE PRECISION
N	SafetyPhIntSec	DOUBLE PRECISION
N	SafetyGrPickupAmps	DOUBLE PRECISION
N	SafetyGrIntSec	DOUBLE PRECISION
N	SwitchGroupId	CHARACTER VARYING(64)
N	WhatIfCode	CHARACTER VARYING(460)
N	DropoutEFuse	INTEGER

Figure 7: InstReclosers Schema

3.2.3 Estimating connected customers at each node

A key requirement of the simulation platform is to provide useful metrics based on the outputs of the FLISR algorithms related to CML and affected customers. As the Synergy data is completely anonymised, it does not contain fine-grained customer related data, therefore it was necessary to derive some approximation of the amount of connected customers by identifying properties that may aid in determining this data point. From analysis of the Synergy output, a table was identified named “Loads”, described in Figure 8 which references connected customers on each phase and links to the “InstSection” table via the “SectionId” property.

By correlating and appending the “Phase<N>Customers” to each “InstSection” it was possible to see an approximation of the connected customers per section of network. The word “approximation” is important to note here however, as specific sections of network had a large number of connected customers while viewing these locations via satellite perspective showed a low number of buildings in the area, one possibility is that these may be industrial, farming or other large customers which are treated differently within the Synergy system. Accounting for these

Loads		
FK,N,U	SectionId	CHARACTER VARYING(64)
N	IsSpotLoad	INTEGER
N	Phase1Customers	DOUBLE PRECISION
N	Phase2Customers	DOUBLE PRECISION
N	Phase3Customers	DOUBLE PRECISION
N	Phase1Kw	DOUBLE PRECISION
N	Phase2Kw	DOUBLE PRECISION
N	Phase3Kw	DOUBLE PRECISION
N	Phase1Kvar	DOUBLE PRECISION
N	Phase2Kvar	DOUBLE PRECISION
N	Phase3Kvar	DOUBLE PRECISION
N	Phase1Kva	DOUBLE PRECISION
N	Phase2Kva	DOUBLE PRECISION
N	Phase3Kva	DOUBLE PRECISION
N	Phase1Kwh	DOUBLE PRECISION
N	Phase2Kwh	DOUBLE PRECISION
N	Phase3Kwh	DOUBLE PRECISION
N	LocationToModelSpotLoads	CHARACTER VARYING(2)
N	WasSetWithAMI	INTEGER
N	CapacityFactor	DOUBLE PRECISION
N	Description	CHARACTER VARYING(200)

Figure 8: Loads Schema

outliers, a total number of customers was added as a property to on pole circuit breaker assets in the data, to allow for the collection of customer and CML related metrics.

3.2.4 Definition of node model

Finally, while the Synergy data includes a unique identifier for each section, node, asset etc., outside of the context of the Synergy system, these identifiers have no meaning and appear to be randomly generated. For the Irish network, assets such as on pole circuit breakers, are identified with a unique SCADA identifier, LBFM switches are denoted with an "S" identifier e.g., S514, with Reclosers denoted with an "R" e.g., R341. From the related research many of these identifiers were available for the Waterford trial site network, for those not available for Kerry or Mullingar, realistic identifiers were assigned following the same format. Likewise, assets that communicate with the DSO SCADA system are given an additional identifier which maps to their SCADA identification, named an ASDU (Application Service Data Unit), again a number of these were available for assets from previous research, those unavailable were given realistic values.

The relevant properties ascertained from the entities included in the Synergy data, supplemented with the approximated customers and SCADA specific identifiers, formed the properties of each object in the graph database, an example is given below for the properties associated with an LBFM switch.

```json
{
    "properties": {
            "SubstationId": "KILCARRAGH_978",
            "ToNodeId": "501685033",
            "PhaseConductorId": "DUMMY",
            "UniqueDeviceId": "501685032",
            "Latitude": 52.234014486994,
            "SectionPhases": "RST ",
            "ConnectedCustomers": {
                    "low": 171,
                    "high": 0
            },
            "Longitude": -7.05518490362683,
            "Name": "Knockboy",
            "Asdu_ca": "24969",
            "SwitchType": "Soule",
            "ScadaId": "S507",
            "IsToEndOpen": {
                    "low": 0,
                    "high": 0
            },
            "FeederId": "PASSAGHE E_978",
            "NearNodeId": "501685033",
            "NodeId": "100000615",
            "FromNodeId": "100000615",
            "SectionId": "501685032"
    }
}
```


All objects in the graph database adhere to the same pattern, whether it is a pole, LBFM switch, Recloser etc., though some of the properties may differ depending on the asset type, this level of consistency negated the need for complex queries and joins thereby enabling a streamlined approach in leveraging the data for use in both the simulation platform and FLISR algorithms.

3.2.5 Network simulation platform

The FLISR Edge simulation platform enables the simulation of fault events which will serve to both test and evaluate the FLISR algorithms in lieu of actual hardware, these simulations are based on the distribution grid topologies obtained from the Synergy GIS output described previously and captured in a graph database format. The simulation platform leverages a microservice architecture, whereby services are loosely coupled, in addition these services are hosted within a containerised environment via “Docker”, and utilise tools such “Docker Compose” and “GNU Make” to enable simple deployment on either local systems or to a remote host [5]. Figure 9 describes the architecture implemented for the simulation platform with a description of each service following.

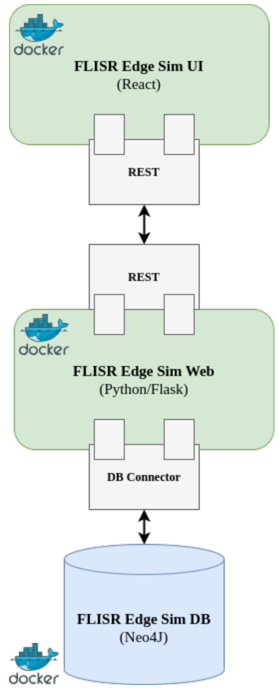


Figure 9: FLISR Edge simulation platform architecture

- **FLISR Edge Sim DB**

The primary database for the simulation platform is hosted within a Docker container built on the "neo4j:5-community" image, this service hosts the modified Synergy data which is ingested via an import script leveraging the Neo4J "Cypher" query language, and is utilised both for network visualisation on the front-end UI and simulation purposes.

- **FLISR Edge Sim Web**

The core logic of the simulation platform is hosted within the "FLISR Edge Sim Web" service, built on the "python:3.11" Docker image, and serves a number of functions. These functions include exposing interfaces to grid topology data for visualisation on the front-end, which are returned in "GeoJSON" format. On deployment of the Sim Web service, grid topology data hosted in the graph database is parsed and a number of ".json" files are generated for grid features, such as single phase lines, three phase lines, on pole circuit breakers and substations, which are treated as "features". By utilising the GeoJSON format, the bulk of the heavy processing is front-loaded on deployment as the front-end simply processes the JSON files returned by the web service, in addition, the GeoJSON format provides for additional properties to be appended to features, which can be used to reference specific assets when simulations are run. An example GeoJSON feature for a switch asset is given below.

```
{
        "type": "Feature",
        "geometry": {
                "type": "Point",
                "coordinates": [52.1755714638251, -7.03211166714884]
        },
        "properties": {
                "sectionId": "100006634",
                "asdu_ca": "24972",
                "scadaId": "S510",
                "switchType": "LBFM",
                "isToEndOpen": "False",
                "name": "Gaultier Kilcarragh",
                "connectedCustomers": "40",
                "feederId": "GAUTIER_978"
        }
}
```

The Web Sim service also includes interfaces to interact with both the centralised and distributed FLISR algorithms for simulation purposes, explained in greater detail in Section 3.2.1, and generating metrics based on the results of completed simulations, returning these results to the user and also capturing them in CSV format.

- **FLISR Edge Sim UI**

Finally, the FLISR Edge Sim UI service provides the front-end visualisation of the distribution grids to the user via browser. This service is built on the "node:19" Docker image and makes use of the "React" framework to implement front-end features. Network lines and assets are visualised via the GeoJSON generated by the FLISR Edge Sim Web service and leveraging the "Leaflet" JavaScript Library in combination with "OpenStreetMaps" through their public tile server. Leaflet is a lightweight, open-source library designed with performance in mind, and given the size of the distribution grids and the amount of lines to be drawn, made it a suitable choice for implementation. Within the Leaflet framework, maps and entities drawn within are treated as separate objects with properties, allowing for these objects to be referenced as needed giving the scope and flexibility to introduce intuitive actions and responses based on user input [7].

3.2.6 Overview of trial site topologies

The distribution grids investigated throughout the research activities and development of the FLISR Edge simulation platform and algorithms, are dispersed throughout Ireland, located in County Waterford, County Kerry and County Westmeath and this section will describe these trial sites in terms of their structure, how their distinct lines are connected and the assets available with circuit breaking equipment installed.

3.2.7 Waterford trial site topology

Of the three trial sites, the distribution grid located in Waterford is by far the most complex, consisting of an intricate mesh network covering the outskirts of Waterford City, following the coastline towards Passage East, Dunmore East and Tramore, also covering the area of countryside between these locations, as presented in Figure 10. This network is fed by three substations, located in Grange, Tramore and Kilcarragh with a total of eighteen on pole circuit breakers installed, consisting of fifteen LBFM switches and three Reclosers. Due to the large area of the site, there are a large number of customers spread throughout, with the highest density occurring within the vicinity of Passage East and Dunmore East as to be expected. Taking into account the sites complexity and proximity to the coastline, there is a higher likelihood of fault events due to inclement weather and as such would be suitable candidate for testing a FLISR solution in the field.

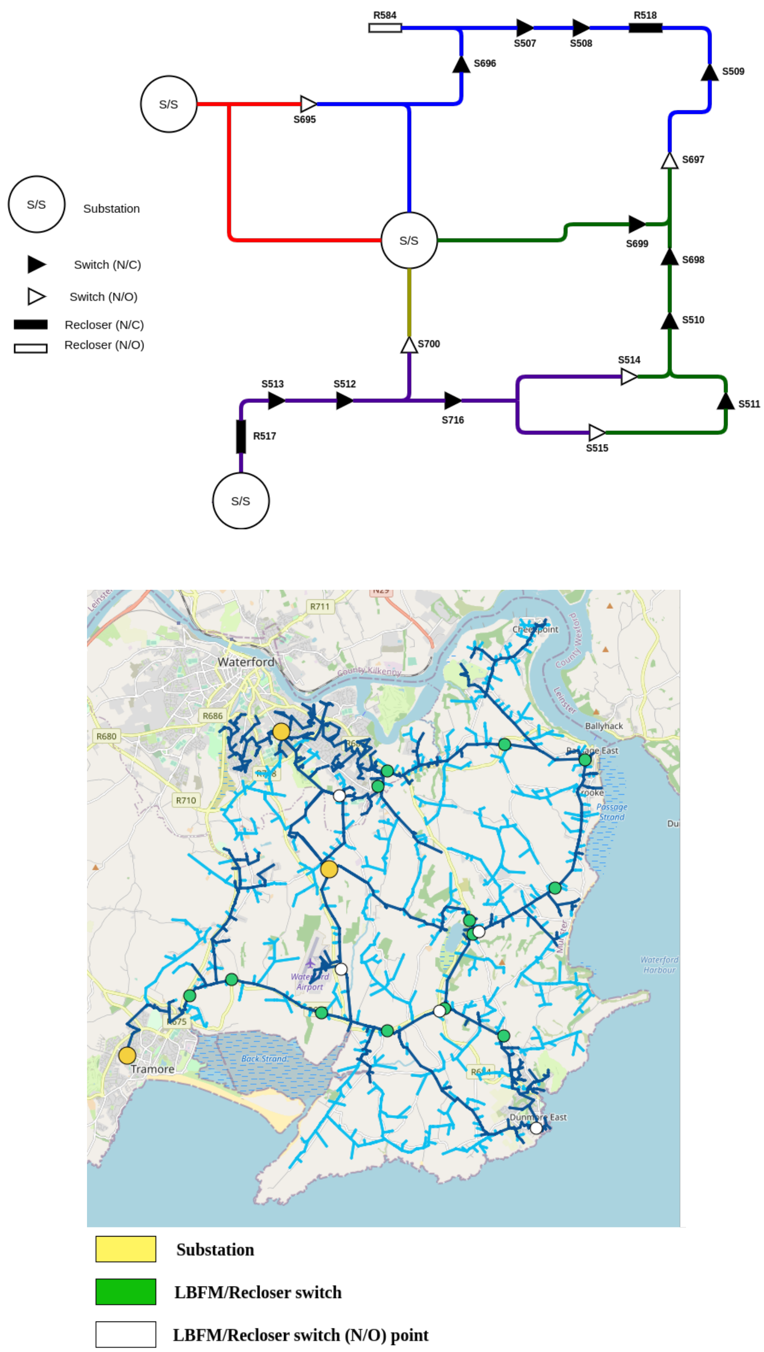


Figure 10: Waterford trial site circuitry and visualisation

3.2.8 Kerry trial site topology

Similar to Waterford, the distribution grid located in Kerry is situated by the coastline, however in comparison the grid is far less complex, fed at both ends by a substation located in Kenmare and Cahersiveen, with a total of eight Recloser switches, the circuitry and visualisation of this network is illustrated in Figure 11. Due to the structure of this network, there is a narrow scope to back feed supply in the event of a fault, and with its remote nature it is likely that network connectivity is limited, particularly in higher areas and mountainous valleys, making it a suitable use case for installation of edge devices to provide distributed FLISR functionality.

Furthermore, as the site is configured with Reclosers in tandem, there is likely a loop automation system in place which provides FLISR-like functionality. For a physical trial, a testing criteria for the FLISR Edge solution would be a comparison for any existing loop automation in place on the network in this area in terms of speed of operation and steps taken by each solution.

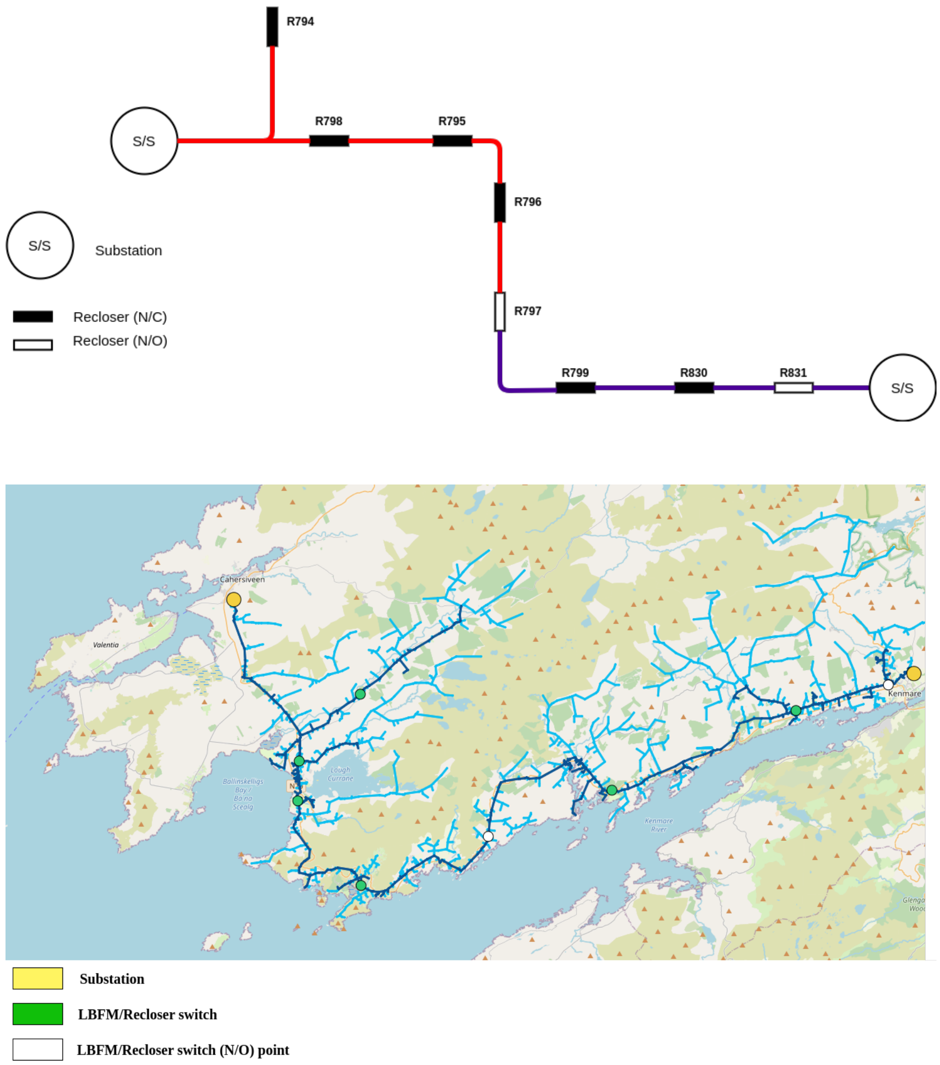


Figure 11: Kerry trial site circuitry and visualisation

3.2.9 Mullingar trial site topology

Lastly, the distribution grid located in Mullingar shares similarities with the previous trial sites, having a somewhat radial structure and similar to the Kerry trial site, configured with Reclosing switches throughout. This network consists of two substations, one located outside of Mullingar town, with the other side of the network fed by a substation located near Clara village to the South with one LBFM and seven Recloser switches installed, Figure 12 presents the circuitry and visualisation of the site. Two of the Reclosers are located at the boundary of the network and serve as normally open points, providing opportunities for back feeding energy supply from adjoining networks. Its location in the midlands likely leads to relatively strong network coverage and a number of fault events that are more representative of the country at large and therefore would serve as an ideal location to test the operation of a FLISR solution in a less unpredictable network.

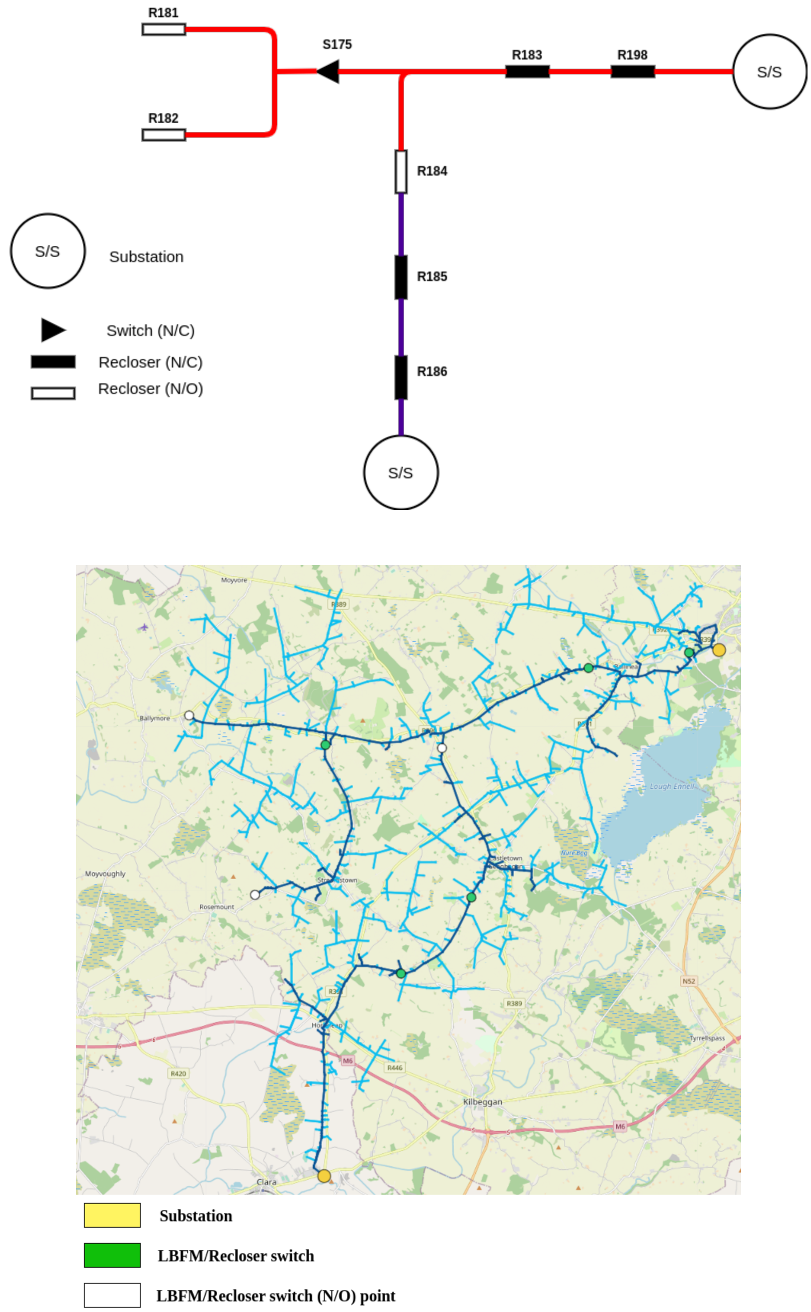


Figure 12: Mullingar trial site circuitry and visualisation

3.2.10 FLISR Edge solution

As described previously, the FLISR Edge solution proposed consists of two applications of the FLISR concept, one of which is a centralised implementation deployed in a remote location, such as the DSOs own systems, with the second deployed on LBFM and Recloser assets by equipping them with an additional edge device, running the distributed algorithm, which can act independently if necessary. This work expands on the previous related research by vastly expanding the capabilities of the FLISR algorithm by introducing bi-directional communication to directly control switching assets, in constrast to the somewhat limited previous iteration which simply generated an email forwarded to DSO personnel [17]. In addition, FLISR Edge introduces the distributed element deployable on a low-cost edge device.

Such an edge device would require compatible interfaces to link with existing equipment on the pole, sufficient processing capability to run Dockerised applications and finally cellular connectivity to send and receive data and control signals. With the rising popularity of low-cost and Edge computing, coupled with the growing rate of processing available to even low-power chips, the options available to provision a low-cost Edge computing device is much higher than in the previous decade [18]. For example, "Raspberry Pi" provide a range of hardware configurations for low-cost computing, including a wide array of additional modules available to meet a variety of use cases. A review of available retailers suggests that on average acquiring a low-cost computing device with the appropriate processing capacity (e.g., Raspberry Pi 3 Model B+) and required modules would cost approximately one hundred euros per device minus installation costs. When considering a network such as Waterford, the cost of retrofitting existing LBFM switches with Reclosers to enable loop automation, costs 61,900 euros per unit according to ESB Networks 2023 standard prices report [22]. In comparison, exploring the application of low-cost Edge computing devices to enable loop automation on existing assets appears more financially feasible.

Assets equipped with switching hardware consist of a number of modules which allow for interfacing with the DSOs SCADA system for interpreting, sending and receiving data and enacting control operations on the physical circuit breaker. At the core of this architecture is a Remote Terminal Unit (RTU), a microprocessor based device, which streams real-time status data to and from the connected circuit breaker [9]. Communication with the DSOs SCADA system conforms to the IEC 60870-5-101/104 protocol, which provides network access based on TCP/IP communications and enables basic control tasks between centralised control centers and connected assets, this is achieved by connecting the RTU to a IEC 101 master, which then connects to a IEC 104 server, facilitating communication via TCP/IP [29]. It is this IEC 104 modem which provides the network communications with the DSO SCADA system, as well as a number of interfaces to connect external devices, including serial and Ethernet connectivity. It is these physical ports that will enable connection with the edge device, allow for the capture and parsing of the switch status messages without interfering with the existing stream to the DSOs SCADA system.

Within the context of the FLISR Edge algorithms operation, the edge device will act as a relay to the centralised FLSIR algorithm when network connectivity is available, by passing switch status messages to the centralised algorithm via MQTT (Message Queuing Telemetry Transport) a lightweight, publish/subscribe bi-directional communication protocol designed for resource constrained devices and/or remote locations [3]. To enable this interaction, an MQTT broker is deployed on the target system (e.g., DSO systems) which connects to the centralised FLISR algorithm, with the edge device exposing an outward MQTT interface. Control messages generated by the centralised FLISR algorithm are captured by the edge device and relayed to the RTU, in the event that remote communication is unavailable, the edge device will instead leverage the local FLISR algorithm and pass the generated control message to the RTU through the IEC 104 connection, outlined in Figure 13.

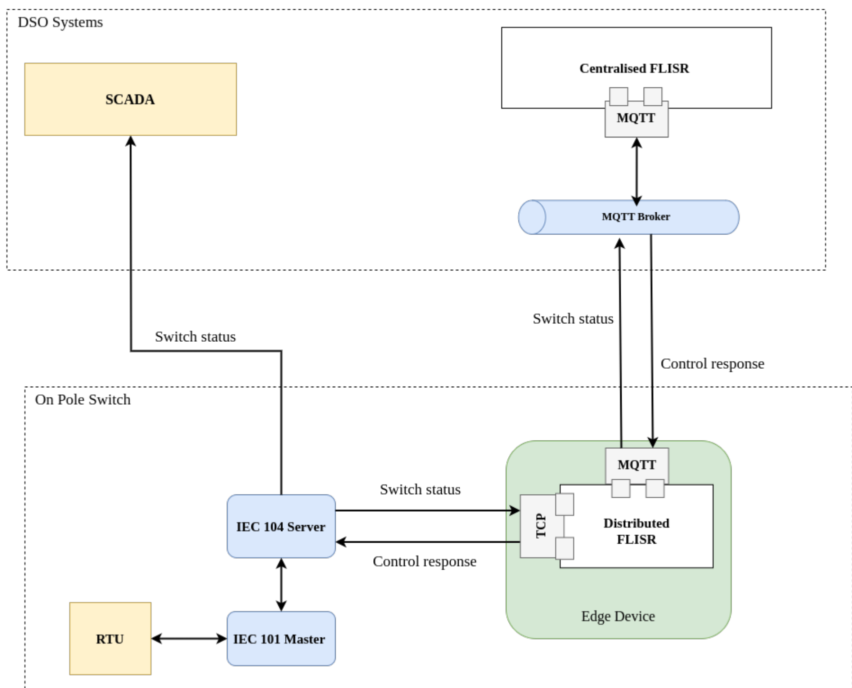


Figure 13: Edge device interaction with switch hardware, DSO system & centralised FLISR algorithm

Switch status and control messages utilised to define actions to take (i.e., opening/closing of the circuit breaker) contain a number of data points, an example is given below in JSON format.


```
{"LinkAdd": 1,"ASDU_CA": 51908,"TypeID": 30,"IOA": 4,"Value": "0(OFF)"}
```


The key properties in this data are, "ASDU_CA" which is the identifier used by the DSO SCADA system to link the switch to its SCADA identifier, "IOA" (Information Object Address) which defines the specific input/output referenced on the switch, for example, the status of the power supply, and finally "Value" which is given as a string value, "0(OFF)" mapping to "False", with a value of "1(ON)" mapping to "True". While LBFM and Recloser switches both follow this convention, the IOAs which map to FLISR specific outputs are distinct as per manufacturer specifications.

As stated in Section 2.2 a switch situated on a damaged line will pass an FPI and LVI status message, whereas a switch which has simply lost power, but not faulted, will pass an LVI status. An FPI on it's own, would not necessarily indicate a fault, for example, a tree branch intermittently knocking against a wire on a gusty day, may result in a FPI status each occurrence, but would not affect supply and be considered a "Transient" fault. As a general rule, any switch which gives an FPI and LVI signal, due to physical damage, should be opened, and therefore the damaged line or lines are no longer "live". The FPI and LVI message format for both a LBFM and Recloser switch are given below in addition to the corresponding control message to open or close the circuit breaker.

- **LBFM switch FPI, LVI & control structure**

The FPI value for a LBFM switch map to IOA 4, meaning a "Fault Current" has been detected.


```
{"LinkAdd": 1,"ASDU_CA": 51908,"TypeID": 30,"IOA": 4,"Value": "1(ON)"}
```


The LVI value maps to IOA 2, which specifies "AC Supply Fail".


```
{"LinkAdd": 1,"ASDU_CA": 51908,"TypeID": 30,"IOA": 2,"Value": "1(ON)"}
```


An IOA value of 8 is used to open ("1(ON)") or close ("0(OFF)") the circuit breaker.


```
{"LinkAdd": 1,"ASDU_CA": 51908,"TypeID": 30,"IOA": 8,"Value": "1(ON)"}
```


- **Recloser switch FPI, LVI & control structure**

The FPI value for a Recloser switch maps to an IOA of 27, specifying a "End of Protection Sequence".


```
{"LinkAdd": 1,"ASDU_CA": 23089,"TypeID": 30,"IOA": 27,"Value": "1(ON)"}
```


The LVI value maps to IOA 34 "Controller Locked Out".


```
{"LinkAdd": 1,"ASDU_CA": 23089,"TypeID": 30,"IOA": 34,"Value": "1(ON)"}
```


Finally, IOA 4096 is utilised to open ("1(ON)") or close ("0(OFF)") the circuit breaker.

```
{"LinkAdd": 1,"ASDU_CA": 23089,"TypeID": 30,"IOA": 4096,"Value": "1(ON)"}
```


In operation, both FLISR algorithms are designed to capture these switch status messages, and based on what is received (FPI and/or LVI) and the topology data related to that pole, whether it is a LBFM/Recloser switch and/or a normally open point, determine the appropriate control message to return for a particular on pole circuit breaker. Similar to the simulation platform services, both the centralised and distributed FLISR algorithms are implemented as lightweight containerised services, and are built on the "golang:1.20-alpine" Docker image. Both algorithms are defined in the Go language, due to its suitability for building lightweight microservices, performance and efficiency. A study by *Dymora et al.* remarked that for processing small amounts of data, Go surpassed comparable languages in terms of performance, CPU and memory utilisation [8], making it an ideal candidate for implementation of the FLISR algorithms where reaction speed is paramount.

As described in Section 1.3 both FLISR algorithms differ in their implementation, the centralised FLISR algorithm is deployed remotely and has access to the complete grid topology hosted within an attached graph database. As such, this implementation of the algorithm will act on a fault event based on the status of all switches on an affected line. Conversely, the distributed FLISR algorithm operates locally, and has access only to the data associated with the specific switch it is monitoring. Figure 14 describes the basic logic of the FLISR algorithm operation when a fault event has been detected.

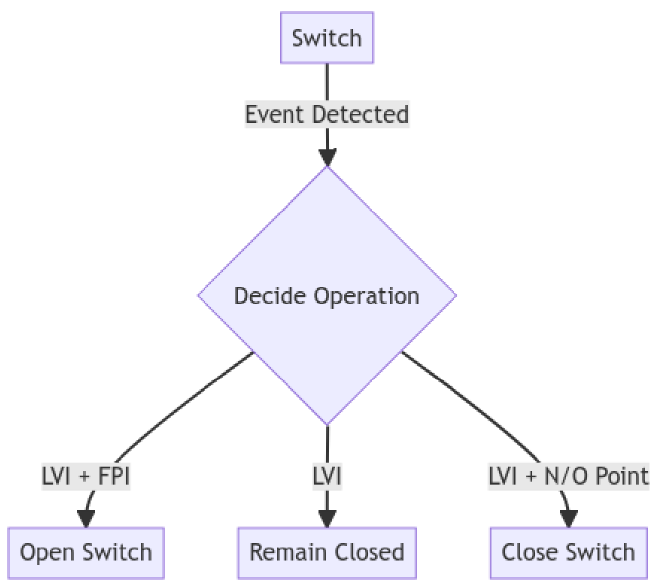


Figure 14: Core FLISR Edge logic

3.3 Experiments

This section describes in detail the experiments undertaken with the simulation platform and FLISR algorithms, how the simulation platform is operated by the user and interacts with the FLISR algorithms, and how the results are captured and presented.

3.3.1 FLISR Edge network simulation platform & algorithm operation

A key component of the FLISR Edge solution is the network simulation test platform. As actual hardware is unavailable to test in the field, the simulation platform serves as an apparatus for demonstrating, testing and validating the FLISR algorithm outputs, in addition to bridging the gap between the research and industry by providing an intuitive tool to DSO personnel. To ensure that the testing is as realistic as possible, the simulation platform implements interfaces to interact with the centralised and distributed FLISR algorithms that mimic those in the field. Furthermore, in order to evaluate the effects of network latency, the centralised algorithm and attached MQTT broker are deployed to a remote VM running within an OpenStack instance, whereas the distributed FLISR algorithm is run locally, as would be the case in the field.

The front-end UI of the network simulation platform is accessed by the user via an internet browser, and presents the three trial sites overlaid on a geographical map populated with single and three phase lines, substations and on pole circuit breakers as presented in Figure 15, and includes menu options to quickly navigate to each trial site.

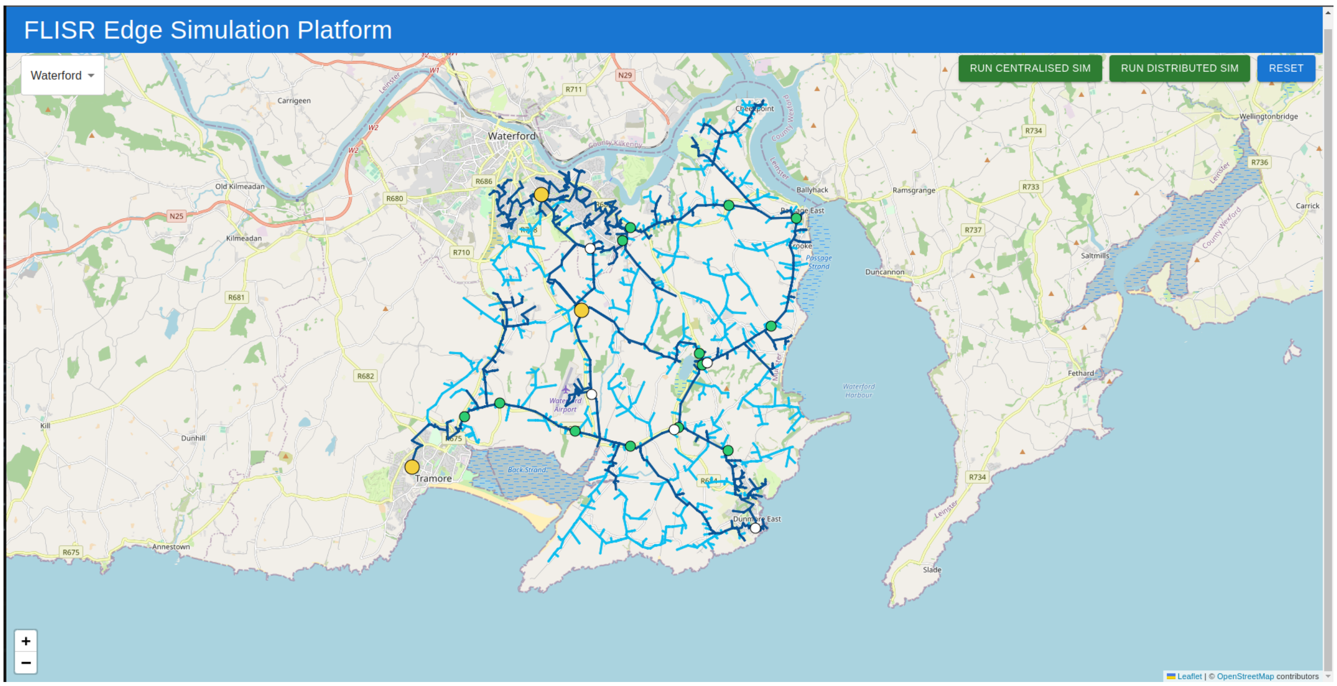


Figure 15: FLISR Edge simulation platform UI

Each switch asset on the map is a clickable object, at which point the user is presented with a popup box containing the properties of the asset, such as the switch type, SCADA identifier, ASDU identifier, whether it is configured as a normally open point and the amount of connected customers on the lines between it and the next switch asset. In addition, each switch has the option to either set it to faulted status, or down status, by selecting a faulted status the simulation platform generates an FPI and LVI status message which corresponds to the switches type (LBFM or Recloser), setting the switch to down status, generates an LVI status message only. Faulted switches are visualised as red markers, whereas down switches are grayed out, and in this way the user can intuitively build a fault case to test as illustrated in Figure 16.

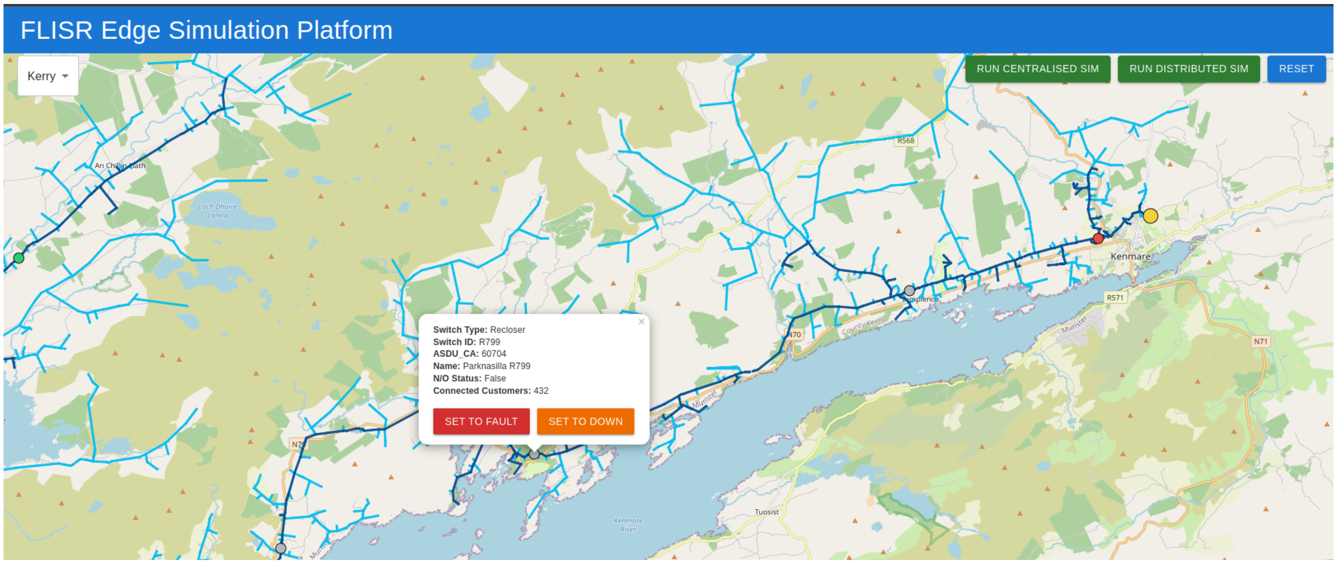


Figure 16: UI FLISR test case example

Once the user is ready to run the test case based on their configuration of faulted and downed assets, they may select to run the test in either centralised or distributed FLISR mode, depending on the option chosen, the FLISR Edge sim web service will create the corresponding FPI and LVI status messages in the JSON formats described in Section 3.1.3 and pass them to the relevant FLISR algorithm. In the case of the centralised FLISR algorithm, the generated FPI and LVI messages are passed via MQTT to the remote broker, each generated control message is passed back on its own unique topic based on the assets ASDU indentifier, as presented in figure Figure 17.

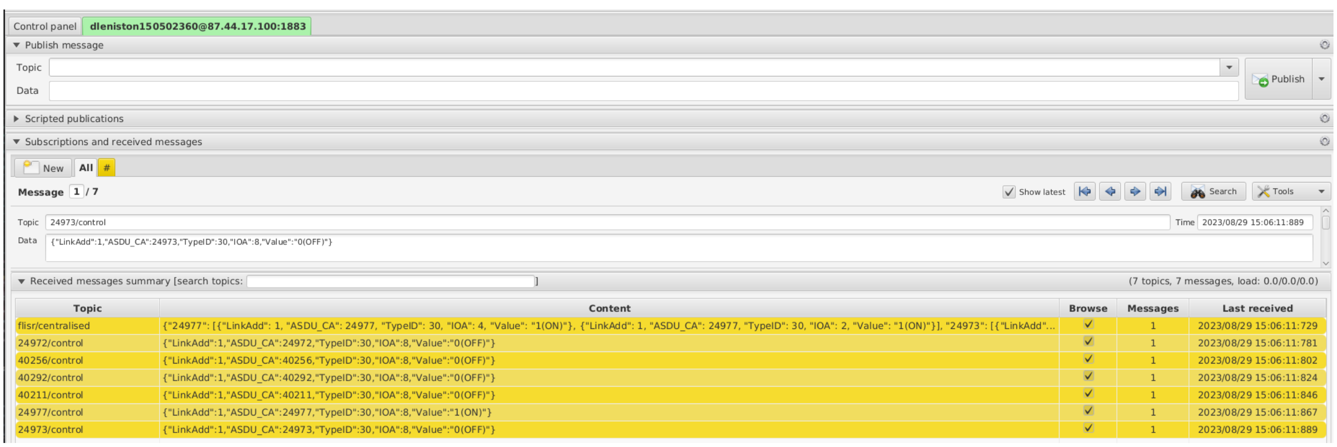


Figure 17: MQTT test case and control response

When running in distributed mode, the FLISR Edge sim web service passes the FPI and LVI messages to the local distributed FLISR algorithm via TCP web socket interface, as would be the case when deployed on the pole via an edge device, in addition to the properties of the pole in the same format as returned from the graph database. This interaction of the simulation platform with the FLISR algorithms is described in Figure 18

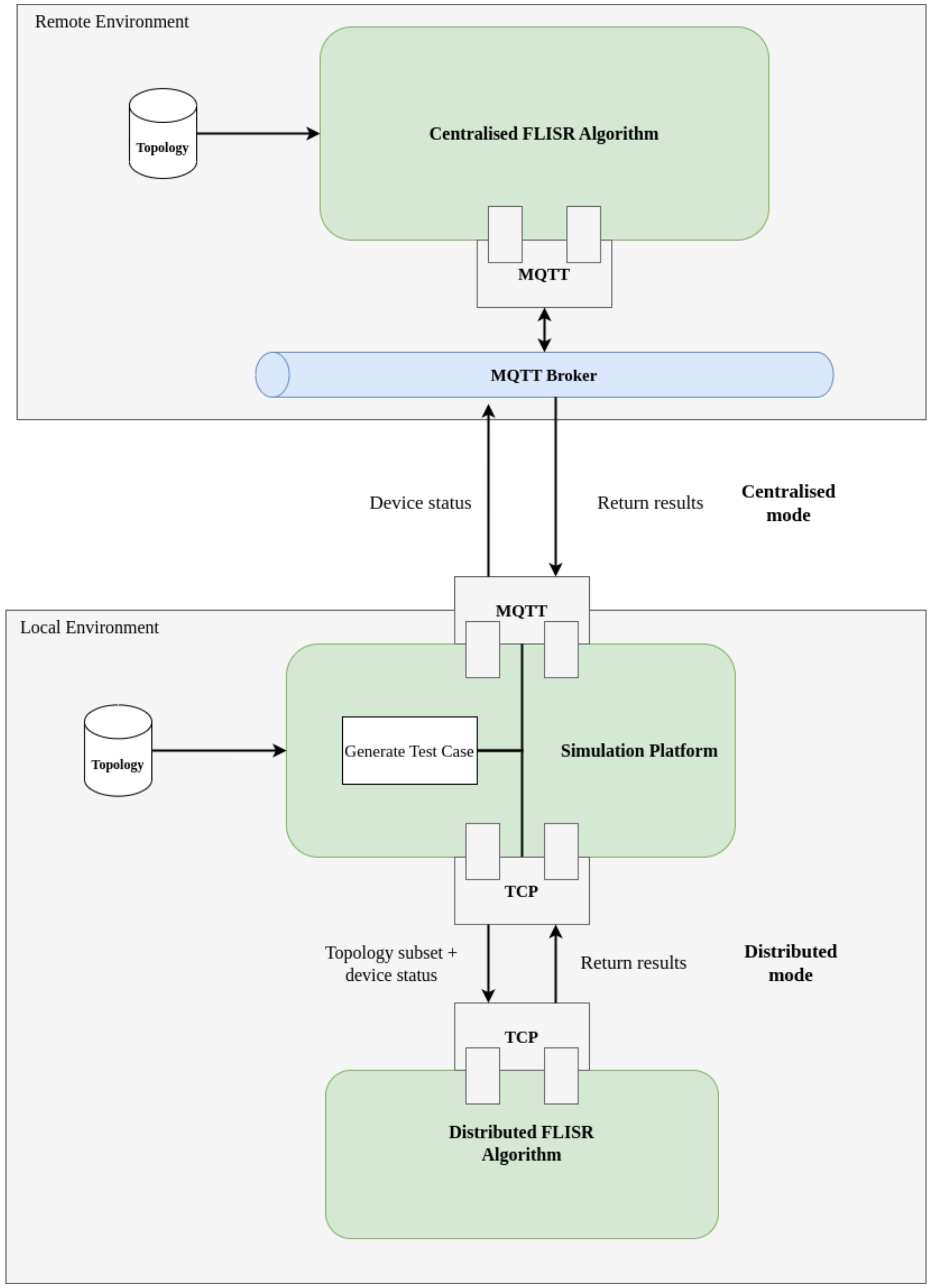


Figure 18: FLISR Edge simulation platform interfaces & interaction

Control messages generated by the FLISR algorithms are captured by the simulation platform, and results are calculated based on a number of key metrics. These metrics include time of operation, which starts before the simulation platform starts polling the FLISR algorithms with the test case, and ends when all control messages are returned for all affected assets. The results also include an overview of which assets were affected as part of the test case, and what control operations were taken as a result of the FLISR algorithms output, the total number of affected customers before and after FLISR operation is given, and based on this figure a calculation of CML per hour before and after the FLISR operation is calculated.

These generated results are returned to the user via the FLISR Edge network simulation platform UI as presented in Figure 19, in addition to being captured in a results CSV file, which is updated with each test case run by the user, which can be used to gather and collate results for a number of test cases for later validation and verification, an example output of the CSV report is presented in Table 2.

Table 2: Distributed FLISR algorithm sample results CSV

start time	end time	faulted devices	down devices	operation	affected customers pre operation	affected customers post operation	cml per hour pre operation	cml per hour post operation
2023-08-25 08:11:37.300	2023-08-25 08:11:37.303	['S513']	"['S512', 'S716', 'S514', 'S700']"	"S513 opened, S512 closed, S716 closed, S514 N/O point closed, S700 N/O point closed, "	627	0	35739	0
2023-08-29 14:31:58.414	2023-08-29 14:31:58.416	['R186']	"['R185', 'R184']"	"R186 opened, R185 closed, R184 N/O point closed, "	250	0	14250	0
2023-08-29 14:35:55.852	2023-08-29 14:35:55.854	"['S507', 'S508']"	"['R518', 'S509', 'S697']"	"S507 opened, S508 opened, R518 closed, S509 closed, S697 N/O point closed, "	1370	171	78090	9747
2023-08-29 14:36:35.252	2023-08-29 14:36:35.255	"['R517', 'S513', 'S512']"	"['S716', 'S514', 'S700']"	"R517 opened, S513 opened, S512 opened, S716 closed, S514 N/O point closed, S700 N/O point closed, "	784	278	44688	15846

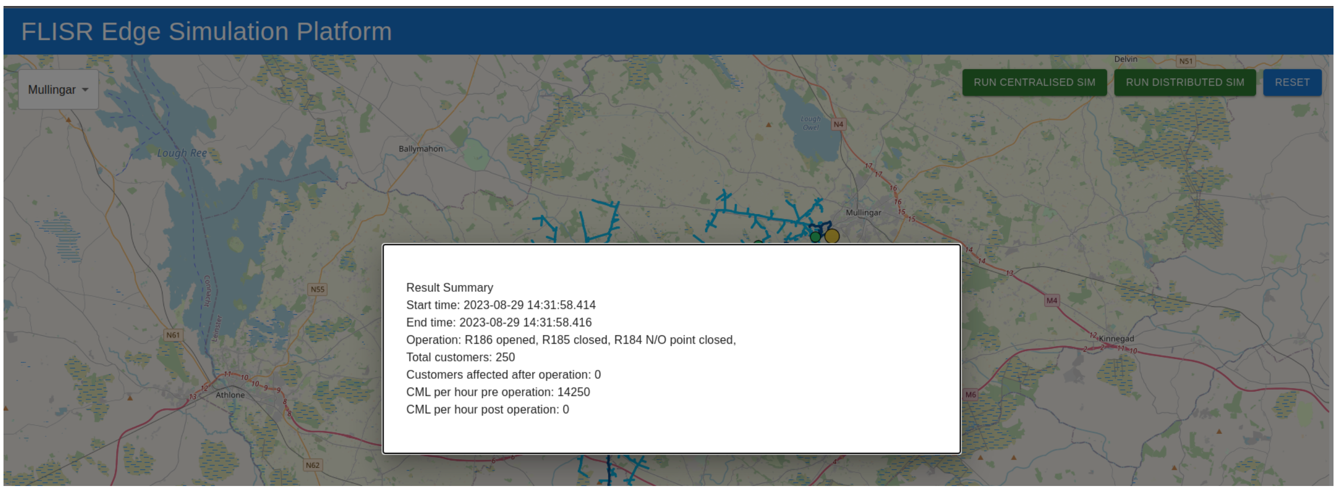


Figure 19: Simulation result from UI perspective

4 Results of the research

The following section presents the results of the experiments conducted and how these relate to the research objectives, details the findings, the relevance of these findings to the DSO and consumer, and proposes future work to expand the solution further.

4.1 Research findings

The findings from the experiments described in Section 3.2.1 are discussed in the context of the research objectives and what insights can be gleaned by framing the results of testing the FLISR Edge solution in terms of these objectives.

- **R1 - How effective will the outputs of the FLISR solution be in terms of reducing DSO fault KPIs?**

From simulating a variety of fault cases utilising the FLISR Edge simulation platform, it has been found that the reduction of DSO fault KPIs, such as CML, varies on a case by case basis. For example, test cases where there are a low vs high occurrence of faulted switch devices, the difference in affected customers and the resulting CML generated per hour can be drastic. Additionally, results for networks which are less complex and options for back

fed supply are limited, such as Kerry for example, the results are less impressive. This observation supports the research literature where it is found that FLISR is most effective on complex, radial networks, such as the network located in Waterford [10].
In addition, it is important to note that FLISR is a fault mitigation solution, therefore in cases where a large number of faulted lines occur on a section of network, the results will be limited. For example, a specific test case simulated for the Waterford trial where one switch is set to faulted, results in 1370 affected customers and 78090 CML per hour prior to FLISR operation, post operation this is reduced to 171 customers affected and 9747 CML per hour. However, increasing the number of faulted switches to four for this same test case, results in an affected customer count of 844 after operation and a total of 48108 CML after FLISR operation.

- **R2 - Given a similar set of conditions, how will the results produced by the centralised FLISR algorithm compare to the distributed FLISR algorithms operating locally at the edge of the network?**

To determine the answer to this question, a wide variety of test cases were simulated across the three distribution grids in centralised and distributed FLISR modes and these were compared and contrasted. From the results gathered, the operational steps generated by the algorithms are identical, with no observable difference in terms of the steps taken to resolve a given fault event. In practice, this is a desirable outcome, as when applied to the field, divergent control operations generated by each algorithm may suggest that the algorithms require refactoring. Furthermore, it is not definitive at this stage that identical results will be given for every situation, as only three networks have been tested in a simulated environment, in addition, with further testing and validation other factors and distribution network features may be found which impact the FLISR algorithm logic.
One key difference observed between the centralised and distributed FLISR algorithms is the operational time. As the centralised algorithm requires a network connection to ingest and output generated control signals, network latency becomes a factor. To better ascertain the impact of network latency on the centralised FLISR operation, an identical test case was simulated ten times on domestic WiFi, analogous to the network bandwidth available on 5G networks, 4G LTE, 3G and 2G, with the average time calculated for each network type as presented in Table 3.

Table 3: Average centralised FLISR operational speed

Network type	Average speed (ms)
Domestic WiFi (5G analogue)	212.8
4G LTE	260.4
3G	267.1
2G	1049.6

From the findings above, the average operational speed even when using low bandwidth networks such as 2G is quite high, this is significant as when applied to the field, the most commonly available networks available are likely to be 3G and 2G, particularly in rural areas. When compared to the distributed FLISR algorithm however, network latency is not a consideration, for the tests conducted the average speed of operation is at or near real-time at two milliseconds.

- **R3 - Given a diverse set of network topologies, will the FLISR solution output valid results for a variety of grid configurations?**

As the findings gathered from the results above have shown, the FLISR Edge solution produces control signals for each of the test cases simulated on the three trial distribution networks. As stated previously, the effectiveness of the solution depends on the network configuration and the level of simulated damage to the grid, for simpler networks such as Kerry and Mullingar, the effectiveness of the solution can be more limited compared to a radial network such as Waterford in terms of reducing the number of affected customers and CML accumulated. However, another aspect of quickly isolating faulted networks is one of safety, where downed lines which are still live pose a safety threat to citizens and therefore the immediate operation enabled by the FLISR Edge solution would assist in alleviating this danger. A more comprehensive collection of results for both algorithms can be found in Appendix C.

4.2 Relevance to the DSO & future work

As described in Section 1.2 the DSO in Ireland is faced with incentives by the energy regulator to reduce the level of service interruption to the consumer, in the form of financial penalties if these targets are not met, and these penalties can be costly, ranging in the millions of euros. Therefore, the benefit of solutions such as FLISR Edge to the DSO are apparent, as the level of damage to the grid and the faults generated thus will increase as unstable weather conditions become more commonplace as climate change worsens. Such solutions also benefit the consumer as energy supply and availability is stabalised and based on the metrics gathered from simulating a range of fault events, the mitigation in terms of customers affected can be quite large. By developing a FLISR implementation which can operate independently on the network via edge computing techniques, FLISR Edge enables a fault-tolerant solution which could be applied to particularly complex networks at a relatively low cost.

The implementation of the network simulation platform also plays a key role in not only demonstrating the operation of the FLISR Edge algorithms but also by bridging the gap between the research and industry by providing a means of showcasing the research to the DSO and leveraging DSO personnel to test, validate and give feedback on the results generated by the FLISR algorithms which can be utilised to continually make adjustments and improvements to the solution. The next stage of development will require this level of interaction from the DSO and the FLISR Edge solution will be leveraged as a use-case for future proposals in the areas of smart energy and edge computing as outlined further in Section 6.

In addition, there are features outlined during the planning process which were not implemented in the current working version of the FLISR Edge solution, due to feasibility issues identified. For example, simulating the flow of energy through the network is not possible to achieve in an accurate way through the grid topology data available at this stage, and would likely require additional data from the DSOs SCADA system to implement, however, these planned features grant a direction to continued development in future research.

Finally, a potential enhancement to the FLISR solution may be feasible through the investigation of transient faults, briefly discussed in Section 3.2.6, by examining the applicability of predictive analysis to determine cases where a fault is likely to occur. Oftentimes, consistent transient faults are a precursor to a sustained fault through physical damage in the future, and by highlighting these occurrences for investigation by the DSO, may allow for preventative steps to be taken such as trimming overgrown tree lines.

5 Conclusion

This document has discussed how the energy network is a vital utility which enables business as usual and daily activities for modern society, and the effect which climate change and inclement weather events have on these activities by interrupting supply to the consumer. Also detailed is the financial penalty this has on the DSO and the motivation to develop and implement solutions which can mitigate the impact of these events and how the FLISR technique is one such solution.

The research undertaken builds on previous related work to meet this need, utilising modern software and communication technologies, in addition to leveraging existing DSO data streams, to provide such a solution in a fault tolerant way by implementing a FLISR application which can operate both remotely and at the edge of the network

by applying edge computing techniques. By automating this process, the time of detection and resolution of fault events are minimised. This grants the potential reduction of the financial impact of fault events, and the level of customers affected. The development of a simulation platform which provides an intuitive interface to create and test fault events on real distribution grids, enables a pathway to interact with DSO personnel to demonstrate, validate and gather feedback to make continual improvements to the FLISR Edge solution.
Such tools are vital in bridging the gap between academic research and industry, and provide a pathway towards continual development, commercialisation and the application of the technologies and techniques explored to other use cases in the energy space, such as ancillary services for grid stability, energy management systems at the industrial and domestic level or interacting with the energy market to enable intelligent energy flexibility to meet the demands on increased RES, distributed energy sources and prosumer activities.

6 Exploitation Activities

Work has been undertaken to explore potential exploitation activities to maximise the outputs of the research, in an effort to gauge the interest in the research topic in the wider community and to gather insights and feedback from third parties and industry experts.

The first exploitation activity was the SETU Computing Department Open Day, which took place on May 26$^{\text{th}}$, the research topic was submitted for presentation and a poster was developed to support this submission. The event will saw participants from a variety of backgrounds, both academic and industry and provided a valuable backdrop to gather feedback in regards to interest in the research topic from those that likely do not have a background in energy.

The second exploitation activity confirmed concerns the Association of Energy Engineers (AEE). AEE is a non-profit professional society with thousands of members across the globe who are actively engaged in the energy sector in both an industrial and scientific context. AEE was originally founded in the US, however the Irish chapter have set a conference to take place in the SETU Arena on the 14$^{\text{th}}$ of September 2023. This research was submitted as an abstract and accepted, and will form one of the keynote presentations of the conference. This will present an invaluable opportunity to demonstrate the final research outputs to members of the energy sector and gather expert feedback to take into consideration.

In addition, the research will be leveraged and expanded to meet a use case for two potential upcoming Horizon Europe proposals in 2024, specifically the call “Resilient Infrastructure 2024 (HORIZON-CL3-2024-INFRA-01)” under the topic Advanced real-time data analysis used for infrastructure resilience and the call “Cross-sectoral solutions for the climate transition (HORIZON-CL5-2024-D2-01)” under the topic Emerging energy technologies for a climate neutral Europe.

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A Appendices

A.0.1 FLISR Edge simulation platform user stories

Table 4: CRO User Stories

User story number	As a ”role”	I want to ”goal/desire”	So that ”benefit”
CRO-01	As a CRO	I want to view a distribution grid on a geographical map	I can see the structure of the distribution grid
CRO-02	As a CRO	I want to view LBFM switches deployed on a distribution grid	I can see the number of on pole circuit breakers installed on a distribution grid
CRO-03	As a CRO	I want to view LBFM metadata	I can read the specifications of selected switches
CRO-04	As a CRO	I want to view distribution grid substations	I can see the location of substations feeding the distribution grid
CRO-05	As a CRO	I want to visualise the flow of energy supply	I can see how the energy supply is proliferating throuout the network
CRO-06	As a CRO	I want to simulate a fault on the grid	I can test the effects a specific fault event will have on the energy supply
CRO-07	As a CRO	I want a simulated fault event to be resolved by a FLISR algorithm	I can see the effect the FLISR algorithm has in response to a simulated fault event

Table 5: DSO User Stories

User story number	As a ”role”	I want to ”goal/desire”	So that ”benefit”
DSO-01	As a DSO	I want access to customer metrics	I have context for the amount of MV and LV customers on a given distribution grid
DSO-02	As a DSO	I to view the CML of a simulated fault event	I can report on the effect a fault event would have on customers
DSO-03	As a DSO	I to view an approximation of the cost of a simulated fault event	I can report on the potential financial penalty of a fault event

A.0.2 FLISR Edge user stories

Table 6: FLISR Master user stories

User story number	As a ”role”	I want to ”goal/desire”	So that ”benefit”
FM-01	As a FLISR Master	I want to have a central view of the distribution grid	I can see the structure of the distribution grid
FM-02	As a FLISR Master	I to have a central FLISR algorithm	I can resolve faults from a central location
FM-03	As a FLISR Master	I want a view of edge devices	I can monitor edge devices installed on the grid
FM-04	As a FLISR Master	I want the ability to provision a network topology subset	FLISR nodes have a subset of the network related to them available
FM-05	As a FLISR Master	I want to provision a distributed FLISR algorithm	FLISR nodes have a subset of the FLISR algorithm
FM-06	As a FLISR Master	I want to recieve a heartbeat from FLISR nodes	I can monitor the connection of FLISR nodes on the grid
FM-07	As a FLISR Master	I want to recieve the status of LBFM switches from FLISR nodes	I know when a fault event occurs
FM-08	As a FLISR Master	I want to send commands to FLISR Nodes	I can action on fault restoration steps

Table 7: FLISR Node user stories

User story number	As a ”role”	I want to ”goal/desire”	So that ”benefit”
FN-01	As a FLISR Node	I want to have a subset of the network topology	I can see the structure of the line on which I am located
FN-02	As a FLISR Node	I want to have a subset of the FLISR algorithm	I can resolve a fault event
FN-03	As a FLISR Node	I want to monitor the status of a switch	I can detect FPI and/or LVI
FN-04	As a FLISR Node	I want to sent a switches status to the FLISR Master	In the event of a fault the status of the switch is sent to the FLISR Master to process
FN-05	As a FLISR Node	I want to send a heartbeat to the FLISR Master	I can determine if communication is available
FN-06	As a FLISR Node	I want to act independently if communication with the FLISR Master is not available	I can use the subset of the FLISR algorithm to operate a switch if communication is not possible

A.0.3 FLISR Edge algorithms results sample

Table 8: Centralised FLISR algorithm results sample

start time	end time	faulted devices	down devices	operation	affected customers pre operation	affected customers post operation	cml per hour pre operation	cml per hour post operation
2023-09-02 13:18:45.519	2023-09-02 13:18:45.726	[’S507’]	”[’S508’, ’R518’, ’S509’, ’S697’]”	”S507 opened, S508 closed, R518 closed, S509 closed, S697 N/O point closed, ”	1370	0	78090	0
2023-09-02 13:27:35.724	2023-09-02 13:27:35.916	[’S507’, ’S508’]	”[’R518’, ’S509’, ’S697’]”	S507 opened, S508 opened, R518 closed, S509 closed, S697 N/O point closed,	1370	171	78090	9747
2023-09-02 13:30:09.237	2023-09-02 13:30:09.413	”[’S507’, ’S508’, ’R518’, ’S509’]”	”[’S697’]”	”S507 opened, S508 opened, R518 opened, S509 opened, S697 N/O point closed, ”	1370	1136	78090	64752
2023-09-02 13:31:06.018	2023-09-02 13:31:06.168	”[’S696’, ’S699’]”	”[’S698’, ’S510’, ’S697’]”	”S696 opened, S699 opened, S698 closed, S510 closed, S697 N/O point closed, ”	152	102	8664	5814
2023-09-02 13:31:27.330	2023-09-02 13:31:27.512	”[’R517’]”	”[’S513’, ’S512’, ’S716’, ’S700’, ’S514’]”	”R517 opened, S513 closed, S512 closed, S716 closed, S700 N/O point closed, S514 N/O point closed, ”	784	0	44688	0
2023-09-02 13:32:02.604	2023-09-02 13:32:02.785	”[’R517’, ’S513’]”	”[’S512’, ’S700’, ’S716’, ’S514’]”	”R517 opened, S513 opened, S512 closed, S700 N/O point closed, S716 closed, S514 N/O point closed, ”	784	157	44688	8949
2023-09-02 13:33:33.993	2023-09-02 13:33:34.173	”[’R517’, ’S513’, ’S512’, ’S716’, ’S700’]”	”[’S514’]”	”R517 opened, S513 opened, S512 closed, S700 N/O point closed, S716 closed, S514 N/O point closed, ”	784	756	44688	43092
2023-09-02 13:34:20.536	2023-09-02 13:34:20.766	”[’S515’, ’S511’]”	”[’S510’, ’S514’, ’S698’, ’S697’, ’S699’]”	”S515 N/O point opened, S511 opened, S510 closed, S514 N/O point closed, S698 closed, S697 N/O point closed, S699 closed, ”	1136	1038	64752	59166
2023-09-02 13:35:04.120	2023-09-02 13:35:04.260	”[’R830’, ’R799’]”	”[’R797’]”	”R830 opened, R799 opened, R797 N/O point closed, ”	812	380	46284	21660
2023-09-02 13:35:51.775	2023-09-02 13:35:51.928	[’R830’]	”[’R799’, ’R797’]”	”R830 opened, R799 closed, R797 N/O point closed, ”	812	0	46284	0
2023-09-02 13:36:31.083	2023-09-02 13:36:31.232	[’R794’]	”[’R798’, ’R795’, ’R796’, ’R797’]”	”R794 opened, R798 closed, R795 closed, R796 closed, R797 N/O point closed, ”	1065	0	60705	0
2023-09-02 13:37:10.572	2023-09-02 13:37:10.718	”[’R794’, ’R798’]”	”[’R795’, ’R796’, ’R797’]”	”R794 opened, R798 opened, R795 closed, R796 closed, R797 N/O point closed, ”	1065	156	60705	8892
2023-09-02 13:37:52.575	2023-09-02 13:37:52.718	”[’R794’, ’R798’, ’R795’]”	”[’R796’, ’R797’]”	”R794 opened, R798 opened, R795 opened, R796 closed, R797 N/O point closed, ”	1065	406	60705	23142
2023-09-02 13:38:31.922	2023-09-02 13:38:32.067	”[’R794’, ’R798’, ’R795’, ’R796’]”	”[’R797’]”	”R794 opened, R798 opened, R795 opened, R796 opened, R797 N/O point closed, ”	1065	724	60705	41268
2023-09-02 13:39:27.791	2023-09-02 13:39:27.975	”[’R183’, ’S175’]”	”[’R184’, ’R181’, ’R182’]”	”R183 opened, S175 opened, R184 N/O point closed, R181 N/O point closed, R182 N/O point closed, ”	426	320	24282	18240
2023-09-02 13:40:31.084	2023-09-02 13:40:31.244	[’R186’]	”[’R185’, ’R184’]”	”R186 opened, R185 closed, R184 N/O point closed, ”	250	0	14250	0
2023-09-02 13:41:05.087	2023-09-02 13:41:05.231	”[’R186’, ’R185’]”	”[’R184’]”	”R186 opened, R185 opened, R184 N/O point closed, ”	250	83	14250	4731

Table 9: Distributed FLISR algorithm results sample

start time	end time	faulted devices	down devices	operation	affected customers pre operation	affected customers post operation	cml per hour pre operation	cml per hour post operation
2023-09-02 13:20:16.366	2023-09-02 13:20:16.368	"['S507', 'S507']"	"['S508', 'R518', 'S509', 'S697', 'S508', 'R518', 'S509', 'S697']"	"S507 opened, S508 closed, R518 closed, S509 closed, S697 N/O point closed, "	2740	171	156180	9747
2023-09-02 13:30:25.304	2023-09-02 13:30:25.306	"['S507', 'S508', 'R518']"	"['S509', 'S697']"	"S507 opened, S508 opened, R518 opened, S509 closed, S697 N/O point closed, "	1370	844	78090	48108
2023-09-02 13:31:45.643	2023-09-02 13:31:45.646	['R517'],	"['S513', 'S512', 'S700', 'S716', 'S514']"	"R517 opened, S513 closed, S512 closed, S700 N/O point closed, S716 closed, S514 N/O point closed, "	784	0	44688	0
2023-09-02 13:32:21.187	2023-09-02 13:32:21.189	"['R517', 'S513']"	"['S512', 'S716', 'S514', 'S700']"	"R517 opened, S513 opened, S512 closed, S716 closed, S514 N/O point closed, S700 N/O point closed, "	784	157	44688	8949
2023-09-02 13:33:51.724	2023-09-02 13:33:51.726	"['R517', 'S513', 'S512', 'S700', 'S716']"	"['S514']"	"R517 opened, S513 opened, S512 opened, S700 N/O point opened, S716 opened, S514 N/O point closed, "	784	479	44688	27303
2023-09-02 13:34:44.674	2023-09-02 13:34:44.677	"['S515', 'S511']"	"['S510', 'S514', 'S698', 'S699', 'S697']"	"S515 N/O point opened, S511 opened, S510 closed, S514 N/O point closed, S698 closed, S699 closed, S697 N/O point closed, "	1136	1038	64752	59166
2023-09-02 13:35:19.620	2023-09-02 13:35:19.622	"['R830', 'R799']"	"['R797']"	"R830 opened, R799 opened, R797 N/O point closed, "	812	380	46284	21660
2023-09-02 13:36:12.840	2023-09-02 13:36:12.841	['R830']	"['R799', 'R797']"	"R830 opened, R799 closed, R797 N/O point closed, "	812	0	46284	0
2023-09-02 13:36:51.611	2023-09-02 13:36:51.614	['R794']	"['R798', 'R795', 'R796', 'R797']"	"R794 opened, R798 closed, R795 closed, R796 closed, R797 N/O point closed, "	1065	0	60705	0
2023-09-02 13:37:30.684	2023-09-02 13:37:30.687	"['R794', 'R798']"	"['R795', 'R796', 'R797']"	"R794 opened, R798 opened, R795 closed, R796 closed, R797 N/O point closed, "	1065	156	60705	8892
2023-09-02 13:38:11.829	2023-09-02 13:38:11.831	"['R794', 'R798', 'R795']"	"['R796', 'R797']"	"R794 opened, R798 opened, R795 opened, R796 closed, R797 N/O point closed, "	1065	406	60705	23142
2023-09-02 13:38:51.291	2023-09-02 13:38:51.293	"['R794', 'R798', 'R795', 'R796']"	"['R797']"	"R794 opened, R798 opened, R795 opened, R796 opened, R797 N/O point closed, "	1065	724	60705	41268
2023-09-02 13:39:52.897	2023-09-02 13:39:52.900	"['R198', 'R183']"	"['R184', 'S175', 'R182', 'R181']"	"R198 opened, R183 opened, R184 N/O point closed, S175 closed, R182 N/O point closed, R181 N/O point closed, "	589	163	33573	9291
2023-09-02 13:40:48.348	2023-09-02 13:40:48.350	['R186']	"['R185', 'R184']"	"R186 opened, R185 closed, R184 N/O point closed, "	250	0	14250	0
2023-09-02 13:41:21.393	2023-09-02 13:41:21.394	"['R186', 'R185']"	"['R184']"	"R186 opened, R185 opened, R184 N/O point closed, "	250	83	14250	4731